

CHAPTER 1

Partial Differential Equations

1.1 Introduction

A differential equation is an equation for a function that relates the values of the function to the values of its derivatives. An ordinary differential equation (ODE) is a differential equation for a function of a single variable, e.g., $x = x(t)$, while a partial differential equation (PDE) is a differential equation for a function of several variables and it is widely used to describe a variety of phenomena such as sound, heat, electrostatics, electrodynamics, fluid flow, elasticity, quantum mechanics and stochastic partial differential equations and so on.

Some examples of PDE are

$$1. \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$$

$$2. \quad \frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2}$$

$$3. \quad \frac{\partial^2 z}{\partial x \partial y} = \left[\frac{\partial z}{\partial x} \right]^2$$

The order of PDE is the order of the highest partial derivative occurring in it. The degree of a PDE is the degree of the highest order partial derivatives occurring in the equation.

In this chapter, we use some standard notations to describe the partial derivatives. Let $z = z(x, y)$ then the following notations are used $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$, $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$. A partial equation by elimination of z is of the form $ax^n + by^n$.

by^n .

1.2 Formation of The Partial Differential Equations

The formation of the partial differential equation from the given relation by eliminating the arbitrary constants or by eliminating the arbitrary functions depending upon whether the given relation involved the arbitrary constants or arbitrary functions.

1.2.1 Formation of The PDE by Eliminating Arbitrary Constants

Working rule :

1. Let the given equation be $f(x, y, z, a, b) = 0$... (1)

has two arbitrary constants.

2. Differentiate (1) partially w.r.to x we get, $\frac{\partial z}{\partial x} = p$... (2)

3. Differentiate (1) partially w.r.to y we get, $\frac{\partial z}{\partial y} = q$... (3)

4. From the equations (1), (2) and (3) eliminate the arbitrary constants, then we get the required PDE.

Note : Some times, the given equation has three arbitrary constants namely a , b and c then we may require to eliminate these three arbitrary constants by using first and second order partial derivatives.

Example 1.2.1. Form the partial differential equation by eliminating the arbitrary constants from $z = ax + by + a^2 + b^2$. **Apr.'11**

Solution . Given equation $z = ax + by + a^2 + b^2$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$a = a. \Rightarrow a = p. \quad \dots (2)$$

$$b = b. \Rightarrow b = q. \quad \dots (3)$$

Using (2) and (3) in (1), we get $z = px + qy + p^2 + q^2$.

Example 1.2.2. Form the partial differential equation by eliminating the arbitrary constants from $z = axy + b$.

Solution . Given equation $z = axy + b$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = ay. \Rightarrow a = \frac{p}{y}. \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = ax. \Rightarrow a = \frac{q}{x}. \quad \dots (3)$$

From equations (2) and (3), we get

$$\frac{p}{y} = \frac{q}{x}. \Rightarrow px = qy \text{ (or) } px - qy = 0.$$

Example 1.2.3. Form the partial differential equation by eliminating the arbitrary constants from $z = ax^2 + by^2$. Nov.'15

Solution . Given equation $z = ax^2 + by^2$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = 2ax. \Rightarrow a = \frac{p}{2x}. \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = 2by. \Rightarrow b = \frac{q}{2y}. \quad \dots (3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} z &= \frac{p}{2x} x^2 + \frac{q}{2y} y^2 = \frac{px}{2} + \frac{qy}{2} \\ \Rightarrow 2z &= px + qy. \end{aligned}$$

Example 1.2.4. Form the partial differential equation by eliminating the arbitrary constants a and b from $z = ax^n + by^n$.

Solution . Given equation $z = ax^n + by^n$.

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = nax^{n-1} \Rightarrow a = \frac{p}{nx^{n-1}} \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = nby^{n-1} \Rightarrow b = \frac{q}{ny^{n-1}} \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$z = \frac{p}{nx^{n-1}} x^n + \frac{q}{ny^{n-1}} y^n = \frac{px}{n} + \frac{qy}{n}$$

$$\Rightarrow nz = px + qy.$$

Example 1.2.5. Form the partial differential equation by eliminating the arbitrary constants $z = xy + y\sqrt{x^2 - a^2} + b$.

Solution . Given equation $z = xy + y\sqrt{x^2 - a^2} + b$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = y + \frac{y}{2\sqrt{x^2 - a^2}} \cdot (2x)$$

$$p = y + \frac{xy}{\sqrt{x^2 - a^2}} \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = x + \sqrt{x^2 - a^2}.$$

$$\Rightarrow q - x = \sqrt{x^2 - a^2} \quad \dots(3)$$

Using (3) in (2), we get

$$p = y + \frac{xy}{q - x}.$$

$$(p - y)(q - x) = xy. \quad (\text{or}) \quad px + qy = pq.$$

Example 1.2.6. Form the partial differential equation by eliminating the arbitrary constants from $(x - a)^2 + (y - b)^2 + z^2 = c^2$. Where c is a fixed constant.

Nov.'11, Apr.'11

Solution . Given equation $(x - a)^2 + (y - b)^2 + z^2 = c^2$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) + 2z.p = 0. \Rightarrow (x - a) = -zp. \quad \dots(2)$$

$$2(y - b) + 2z \cdot q = 0. \Rightarrow (y - b) = -zq. \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} (-zp)^2 + (-zq)^2 + z^2 &= c^2. \\ \Rightarrow z^2(p^2 + q^2 + 1) &= c^2. \end{aligned}$$

Example 1.2.7. Form the partial differential equation by eliminating arbitrary constants from $z = (x - a)^2 + (y - b)^2 + 1$. Nov.'15

Solution . Given equation $z = (x - a)^2 + (y - b)^2 + 1$(1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = 2(x - a). \Rightarrow (x - a) = \frac{p}{2}. \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = 2(y - b). \Rightarrow (y - b) = \frac{q}{2}. \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} z &= \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 + 1 \\ \Rightarrow 4z &= p^2 + q^2 + 4. \end{aligned}$$

Example 1.2.8. Form the partial differential equation by eliminating the arbitrary constants from $z = (x - a)(y - b)$. Apr.'14

Solution . Given equation $z = (x - a)(y - b)$(1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned} \frac{\partial z}{\partial x} &= p = 1 \cdot (y - b). \Rightarrow (y - b) = \frac{p}{1} = p. \\ \frac{\partial z}{\partial y} &= q = 1 \cdot (x - a). \Rightarrow (x - a) = \frac{q}{1} = q. \end{aligned}$$

Using in (1) we get $z = qp$.

Example 1.2.9. Form the partial differential equation by eliminating the arbitrary constants from $z = (x + a)(y + b)$.

Solution . Given equation $z = (x + a)(y + b)$.

...(1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = (y + a) \quad \text{and} \quad \frac{\partial z}{\partial y} = q = (x + b).$$

Using in (1) we get $z = qp$.

Example 1.2.10. Form the partial differential equation by eliminating arbitrary constants from $z = (x^2 + a^2)(y^2 + b^2)$.

Apr.'12, Apr.'15

...(1)

Solution . Given equation $z = (x^2 + a^2)(y^2 + b^2)$.

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned} \frac{\partial z}{\partial x} = p &= 2x.(y^2 + b^2). \Rightarrow (y^2 + b^2) = \frac{p}{2x}. \\ \frac{\partial z}{\partial y} = q &= 2y.(x^2 + a^2). \Rightarrow (x^2 + a^2) = \frac{q}{2y}. \end{aligned}$$

$$\begin{aligned} \text{Using in (1), we get } z &= \left(\frac{q}{2y}\right) \left(\frac{p}{2x}\right) \\ \Rightarrow pq &= 4xyz. \end{aligned}$$

Example 1.2.11. Form the partial differential equation by eliminating the arbitrary constants from $\log(az - 1) = x + ay + b$.

Solution . Given equation $\log(az - 1) = x + ay + b$.

...(1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{1}{(az - 1)} \cdot ap = 1. \Rightarrow ap = (az - 1). \therefore a = \frac{1}{z - p}. \dots(2)$$

$$\frac{1}{(az - 1)} \cdot aq = a. \Rightarrow q = (az - 1). \dots(3)$$

Using (2) in (3) we get

$$\begin{aligned} q &= \left(\frac{z}{z - p} - 1\right) = \frac{p}{z - p} \\ \therefore q(z - p) &= p. \quad (\text{or}) \quad p(q + 1) = zq. \end{aligned}$$

1.2.2 Formation of The PDE by Eliminating Arbitrary Functions

Working rule for eliminating an arbitrary function

1. Let the given equation is of the form $z = f(x, y)$... (1)

Where f is an arbitrary function.

2. Differentiate (1) partially w.r.to x , $\frac{\partial z}{\partial x} = p = f'(x, y)$... (2)

3. Differentiate (1) partially w.r.to y , $\frac{\partial z}{\partial y} = q = f'(x, y)$... (3)

4. From the equations (1), (2) and (3) eliminate the arbitrary function and its derivative, then we get the required PDE.

Example 1.2.22. Form the partial differential equation by eliminating arbitrary function from $z = f(2x + 3y)$.

Solution . Given function $z = f(2x + 3y)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned} \frac{\partial z}{\partial x} &= p = f' \cdot 2 \quad [\because f' = f'(2x + 3y)] \\ \Rightarrow f' &= \frac{p}{2} \end{aligned} \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = f' \cdot 3 \Rightarrow f' = \frac{q}{3} \quad \dots (3)$$

From equations (2) and (3) we have

$$\frac{p}{2} = \frac{q}{3} \Rightarrow 3p - 2q = 0.$$

Example 1.2.23. Form the partial differential equation by eliminating arbitrary function from $z = f(x^2 - y^2)$. Apr.'11, Apr.'12

Solution . Given function $z = f(x^2 - y^2)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned} \frac{\partial z}{\partial x} &= p = f' \cdot 2x \quad [\because f' = f'(x^2 - y^2)] \\ \Rightarrow f' &= \frac{p}{2x} \end{aligned} \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = f' \cdot (-2y) \Rightarrow f' = -\frac{q}{2y} \quad \dots(3)$$

From equations (2) and (3) we have

$$\frac{p}{2x} = -\frac{q}{2y} \Rightarrow 2yp + 2xq = 0. \quad (\text{or}) \quad yp + xq = 0.$$

Example 1.2.24. Form the partial differential equation by eliminating arbitrary function f from $z = f(x^2 + y^2 + z^2)$. Nov.'15

Solution . Given function $z = f(x^2 + y^2 + z^2)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = f' [2x + 2zp] \quad [\because f' = f'(x^2 + y^2 + z^2)] \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = f' [2y + 2zq] \quad [\because f' = f'(x^2 + y^2 + z^2)] \quad \dots(3)$$

$$\frac{(2)}{(3)} \Rightarrow \frac{p}{q} = \frac{f' [2x + 2zp]}{f' [2y + 2zq]} = \frac{(x + zp)}{(y + zq)}$$

$$p(y + zq) = q(x + zp). \Rightarrow py - qx = 0.$$

Which is the required PDE.

Example 1.2.25. Form the partial differential equation by eliminating arbitrary function from $z = f\left(\frac{y}{x}\right)$ and $z = f\left(\frac{x}{y}\right)$

Solution . Given function $z = f\left(\frac{y}{x}\right)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = f' \cdot \left(-\frac{y}{x^2}\right) \quad [\because f' = f'\left(\frac{y}{x}\right)]$$

$$\text{i.e., } p = -\frac{y}{x^2} \cdot f' \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = f' \cdot \left(\frac{1}{x}\right) \quad \dots(3)$$

$$\frac{2}{3} \Rightarrow \frac{p}{q} = -\frac{\frac{y}{x^2} \cdot f'}{f' \cdot \left(\frac{1}{x}\right)} = -\frac{y}{x}$$

$$\therefore px = -qy \quad (\text{or}) \quad xp + yq = 0.$$

Which is the required PDE.

Similarly the required PDE for $z = f\left(\frac{x}{y}\right)$ is $yp + xq = 0$.

Example 1.2.26. Form the partial differential equation by eliminating arbitrary function from $z = f\left(\frac{xy}{z}\right)$.

Solution . Given function $z = f\left(\frac{xy}{z}\right)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned}\frac{\partial z}{\partial x} &= f' \cdot \frac{\partial}{\partial x} \left[\frac{xy}{z} \right] \quad \left[\because f' = f' \left(\frac{xy}{z} \right) \right] \\ \therefore p &= f' \cdot \left[\frac{z \cdot y - xy \cdot p}{z^2} \right] \quad \dots (2)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial y} &= f' \cdot \frac{\partial}{\partial y} \left[\frac{xy}{z} \right] \quad \left[\because f' = f' \left(\frac{xy}{z} \right) \right] \\ \therefore q &= f' \cdot \left[\frac{z \cdot x - xy \cdot q}{z^2} \right] \quad \dots (3)\end{aligned}$$

$$\frac{(2)}{(3)} \Rightarrow \frac{p}{q} = \frac{f' \cdot \left[\frac{zy - xyp}{z^2} \right]}{f' \cdot \left[\frac{zx - xyq}{z^2} \right]} = \frac{zy - xyp}{zx - xyq}$$

$$\text{i.e., } px(z - yq) = qy(z - xp) \quad (\text{or}) \quad px = qy.$$

Which is the required PDE.

Example 1.2.27. Form the partial differential equation by eliminating arbitrary function from $z = e^{ax+by} f(ax+by)$.

Solution . Given function $z = e^{ax+by} f(ax+by)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned}\frac{\partial z}{\partial x} &= ae^{ax+by} f + e^{ax+by} f' \cdot a \quad \left[\because f' = f'(ax+by) \right] \\ \text{i.e., } p &= ae^{ax+by} (f + f'). \quad \dots (2)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial y} &= be^{ax+by} f + e^{ax+by} f' \cdot b \quad \left[\because f' = f'(ax+by) \right] \\ \text{i.e., } q &= be^{ax+by} (f + f'). \quad \dots (3)\end{aligned}$$

$$\begin{aligned}\frac{(2)}{(3)} \Rightarrow \frac{p}{q} &= \frac{ae^{ax+by} (f + f')}{be^{ax+by} (f + f')} = \frac{a}{b} \\ \therefore bp &= aq \quad (\text{or}) \quad bp - aq = 0.\end{aligned}$$

1.2.2.1 Eliminating an Arbitrary Function ϕ from The Form of $\phi(u, v) = 0$

Given function is of the form $\phi(u, v) = 0$... (1)

Where u and v are functions of x and y and z .

Differentiating (1) partially w.r.to x and y we get

$$\frac{\partial \phi}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial x} = 0 \quad \dots (2)$$

$$\frac{\partial \phi}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial y} = 0 \quad \dots (3)$$

Now eliminating ϕ between (2) and (3) we get the required partial differential equation as follows

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

Example 1.2.34. Form the partial differential equation by eliminating the arbitrary function from $f(xy + z^2, x + y + z) = 0$

Nov.'10, Nov.'11

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = xy + z^2$, $v = x + y + z$.

Then $\frac{\partial u}{\partial x} = y + 2z p$ and $\frac{\partial u}{\partial y} = x + 2z q$

Also $\frac{\partial v}{\partial x} = 1 + p$ and $\frac{\partial v}{\partial y} = 1 + q$

Thus the elimination of f becomes,

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} y + 2z p & 1 + p \\ x + 2z q & 1 + q \end{vmatrix} = 0$$

$$(y + 2z p)(1 + q) - (x + 2z q)(1 + p) = 0$$

$$y + 2z p + yq + 2z pq - [x + 2z q + xp + 2z pq] = 0$$

$$y + 2z p + yq - x - 2z q - xp = 0$$

$$(2z - x)p + (y - 2z)q = x - y$$

Example 1.2.35. Form the partial differential equation by eliminating arbitrary function 'f' from $f(z - xy, x^2 + y^2) = 0$

Nov.'10, Nov.'11, Apr.'13

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = z - xy, v = x^2 + y^2$.

Then $\frac{\partial u}{\partial x} = p - y$ and $\frac{\partial u}{\partial y} = q - x$. Also $\frac{\partial v}{\partial x} = 2x$ and $\frac{\partial v}{\partial y} = 2y$

Thus the elimination of f is,
$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} p - y & 2x \\ q - x & 2y \end{vmatrix} = 0$$

$$2y(p - y) - 2x(q - x) = 0$$

$$y(p - y) - x(q - x) = 0$$

$$yp - xq = y^2 - x^2.$$

Example 1.2.36. Form the partial differential equation by eliminating f from $f(x^2 + y^2 + z^2, x + y + z) = 0$. Apr.'09

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = x^2 + y^2 + z^2, v = x + y + z$.

Then $\frac{\partial u}{\partial x} = 2x + 2z p$ and $\frac{\partial u}{\partial y} = 2y + 2z q$. $\frac{\partial v}{\partial x} = 1 + p$ and $\frac{\partial v}{\partial y} = 1 + q$

Thus the elimination of f is,
$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} 2x + 2z p & 1 + p \\ 2y + 2z q & 1 + q \end{vmatrix} = 0$$

$$(2x + 2z p)(1 + q) - (2y + 2z q)(1 + p) = 0$$

$$(x + z p)(1 + q) - (y + z q)(1 + p) = 0$$

$$x + z p + xq + zpq - [y + z q + yp + zpq] = 0$$

$$x + z p + xq - [y + z q + yp] = 0$$

$$(z - y)p + (x - z)q = y - x.$$

Example 1.2.37. Form the partial differential equation by eliminating arbitrary function 'f' from $f(z^2 - xy, \frac{x}{z}) = 0$

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = z^2 - xy$, $v = \frac{x}{z}$.

Then $\frac{\partial u}{\partial x} = 2zp - y$ and $\frac{\partial u}{\partial y} = 2zq - x$. $\frac{\partial v}{\partial x} = \frac{z - xp}{z^2}$ and $\frac{\partial v}{\partial y} = -\frac{xq}{z^2}$

$$\text{Thus the elimination of } f \text{ is, } \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} 2zp - y & \frac{z - xp}{z^2} \\ 2zq - x & -\frac{xq}{z^2} \end{vmatrix} = 0$$

$$(2zp - y) \left(-\frac{xq}{z^2}\right) - \left(\frac{z - xp}{z^2}\right) (2zq - x) = 0$$

Multiply through out by z^2 , we get

$$(2zp - y) (-xq) - (z - xp) (2zq - x) = 0$$

$$-2xzp q + xyq - (2z^2 q - 2zxp q - xz + x^2 p) = 0$$

$$xyq - (2z^2 q - xz + x^2 p) = 0$$

$$x^2 p + (xy - 2z^2) q = xz.$$

Example 1.2.38. Form the partial differential equation by eliminating by eliminating f from $f(\frac{y}{x}, x^2 + y^2 + z^2) = 0$.

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = \frac{y}{x}$, $v = x^2 + y^2 + z^2$.

Then $\frac{\partial u}{\partial x} = -\frac{y}{x^2}$ and $\frac{\partial u}{\partial y} = \frac{1}{x}$. $\frac{\partial v}{\partial x} = 2x + 2z p$ and $\frac{\partial v}{\partial y} = 2y + 2z q$

$$\text{Thus the elimination of } f \text{ is, } \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} -\frac{y}{x^2} & 2x + 2z p \\ \frac{1}{x} & 2y + 2z q \end{vmatrix} = 0$$

$$\left(-\frac{y}{x^2}\right) (2y + 2z q) - \left(\frac{1}{x}\right) (2x + 2z p) = 0$$

$$\begin{aligned}\left(-\frac{y}{x^2}\right)(y + zq) + \left(\frac{1}{x}\right)(x + zp) &= 0 \\ -y(y + zq) + x(x + zp) &= 0 \\ -y^2 - yzq + x^2 + xzp &= 0 \\ xzp - yzq &= y^2 - x^2.\end{aligned}$$

Exercise

Form PDE by eliminating arbitrary function 'f' from:

1. $f(x^2 + y^2, y^2 + z^2) = 0$ **Ans.** $yzp - xzq = xy$.

2. $f(xyz, x + y + z) = 0$ **Ans.** $x(y - z)p + y(z - x)q = z(x - y)$.

3. $f(xyz, x^2 + y^2 + z^2) = 0$ **Ans.** $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.

4. $f(x^2 + y^2 + z^2, x + y + z) = 0$ **Apr.'09**
Ans. $(y - z)p + (z - x)q = (x - y)$.

5. $f(x^2 + y^2 + z^2, lx + my + nz) = 0$ **Nov.'03**
Ans. $(ny - mz)p + (lz - nx)q = (mx - ly)$.

6. $f(x^2 + y^2 + z^2, xy + z) = 0$ **Ans.** $(y + zx)p - (x - yz)q = x^2 - y^2$

7. $f(x^2 + y^2 + z^2, \frac{y}{z}) = 0$ **Ans.** $(y^2 + z^2)p - xyq + xz = 0$.

8. $f(x + y + z, x^2 + y^2 - z^2) = 0$ **Ans.** $(y + z)p - (z + x)q = x - y$.

9. $xyz = \phi(x^2 + y^2 + z^2)$ **Ans.** $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.

[Hint : We may assume the above problem as $\phi(xyz, x^2 + y^2 + z^2) = 0$ then solve it.]

10. $f(\frac{y}{x}, x^2 + y^2 + z^2) = 0$ **Ans.** $xzp + yzq + x^2 + y^2 = 0$

1.2.2.2 Eliminating Two Arbitrary Functions

Note : Some times, the given equation has two arbitrary functions say f and g then we may require to eliminate these three arbitrary functions and their derivatives by using first and second order partial derivatives. i.e., $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$, $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$ and $t = \frac{\partial^2 z}{\partial y^2}$.

Example 1.2.39. Form the partial differential equation by eliminating arbitrary function from the relation $z = f(2x + 3y) + g(2x + y)$.
Nov.'12

Solution . Given relation $z = f(2x + 3y) + g(2x + y)$.

$$\frac{\partial z}{\partial x} = p = 2f'(2x + 3y) + 2g'(2x + y)$$

$$\frac{\partial z}{\partial y} = q = 3f'(2x + 3y) + g'(2x + y)$$

$$\frac{\partial^2 z}{\partial x^2} = r = 4f''(2x + 3y) + 4g''(2x + y)$$

$$\frac{\partial^2 z}{\partial y \partial x} = s = 6f''(2x + 3y) + 2g''(2x + y)$$

$$\frac{\partial^2 z}{\partial y^2} = t = 9f''(2x + 3y) + g''(2x + y)$$

Eliminating $f''(2x + 3y)$, $g''(2x + y)$ from the above last three equations, we have

$$\begin{vmatrix} 4 & 4 & r \\ 6 & 2 & s \\ 9 & 1 & t \end{vmatrix} = 0$$

$$4(2t - s) - 4(6t - 9s) + r(6 - 18) = 0$$

$$-12r + 32s - 16t = 0$$

$$3r - 8s + 4t = 0.$$

Example 1.2.40. Form the partial differential equation by eliminating arbitrary functions from $z = xf(ax + by) + g(ax + by)$. Nov.'09

Solution . Replace $a = 1$ and $b = 1$ in the previous problem and proceed as in the above, finally we get $r - 2s + t = 0$.

Example 1.2.42. Form partial differential equation by eliminating f and ϕ from $z = f(x^2 + y) + \phi(x^2 - y)$. Apr.'14

Solution . Given $z = f(x^2 + y) + \phi(x^2 - y) = f + g$ (say).

$$\frac{\partial z}{\partial x} = p = 2x \cdot f' + 2x \cdot \phi' = 2x[f' + \phi']$$

$$\frac{\partial z}{\partial y} = q = f' - \phi'$$

$$\frac{\partial^2 z}{\partial x^2} = r = 2[f' + \phi'] + 2x[2x f'' + 2x \phi'']$$

$$\text{i.e., } 2[f' + \phi'] + 4x[f'' + \phi''] = r \quad \dots(1)$$

$$\frac{\partial^2 z}{\partial y \partial x} = s = 2x[f'' + \phi''] \quad \dots(2)$$

$$\frac{\partial^2 z}{\partial y^2} = t = f'' + \phi'' \quad \dots(3)$$

Substituting (3) and (2) we get $s = 2xt$.

Example 1.2.43. Form partial differential equation by eliminating f and g from $z = f(x^3 + 2y) + g(x^3 - 2y)$.

Solution . Given $z = f(x^3 + 2y) + g(x^3 - 2y) = f + g$ (say).

$$\frac{\partial z}{\partial x} = p = 3x^2 \cdot f' + 3x^2 \cdot g'$$

$$p = 3x^2[f' + g'] \quad \dots(1)$$

$$\frac{\partial z}{\partial y} = q = 2f' - 2g' = 2[f' - g'] \quad \dots(2)$$

$$\frac{\partial^2 z}{\partial x^2} = r = 6x[f' + g'] + 3x^2[3x^2 f'' + 3x^2 g'']$$

$$\therefore r = 6x[f' + g'] + 9x^4[f'' + g''] \quad \dots(3)$$

$$\frac{\partial^2 z}{\partial y \partial x} = s = 6x^2[f'' - g''] \quad \dots(4)$$

$$\frac{\partial^2 z}{\partial y^2} = t = 4[f'' + g''] \quad \dots(5)$$

Substituting (1) and (5) in (3) we get

$$r = 6x \frac{p}{3x^2} + 9x^4 \frac{t}{4}$$

$$r = \frac{2p}{x} + \frac{9x^4 t}{4}$$

Example 1.2.44. Form partial differential equation by eliminating f and g from $z = x^2 f(y) + y^2 g(x)$.

Solution . Given $z = x^2 f(y) + y^2 g(x) = x^2 f + y^2 g$ (say).

$$\frac{\partial z}{\partial x} = p = 2x f + y^2 g' \quad \dots(1)$$

$$\frac{\partial z}{\partial y} = q = x^2 f' + 2y g \quad \dots(2)$$

$$\frac{\partial^2 z}{\partial x^2} = r = 2f + y^2 g'' \quad \dots(3)$$

$$\frac{\partial^2 z}{\partial y \partial x} = s = 2x f' + 2y g' \quad \dots(4)$$

$$\frac{\partial^2 z}{\partial y^2} = t = x^2 f'' + 2g'' \quad \dots(5)$$

$$\begin{aligned} px + qy &= 2x^2 f + xy^2 g' + x^2 y f' + 2y^2 g \\ &= [2x^2 f + 2y^2 g] + xy^2 g' + x^2 y f' \\ &= 2[x^2 f + y^2 g] + xy[y g' + x f'] \\ &= 2z + xy \left[\frac{s}{2} \right]. \end{aligned}$$

Example 1.2.45. Form the partial differential equation by eliminating the arbitrary functions f and g from $z = f(y/x) + yg(x)$.

Solution . Given $z = x f(y/x) + yg(x) \quad \dots(1)$

Eliminating f and g from (1), So that differentiate partially (1) w.r.to x and y respectively, we get

$$\frac{\partial z}{\partial x} = f(y/x) + x f'(y/x) \cdot \left(-\frac{y}{x^2} \right) + yg'(x)$$

$$\therefore \frac{\partial z}{\partial x} = f(y/x) - \frac{y}{x} f'(y/x) + y g'(x) \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = x f'(y/x) \cdot \frac{1}{x} + g(x) = f'(y/x) + g(x) \quad \dots(3)$$

Finding second order partial derivatives from the above two equations, we have

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{y}{x^2} \cdot f''(y/x) + g'(x) \quad \dots(4)$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{1}{x} f''(y/x) \quad \dots(5)$$

Here the elimination the functions f and g are not possible. Thus we find one more derivative. i.e., Third derivative.

$$\frac{\partial^2 z}{\partial y^2 \partial x} = -\frac{1}{x^2} f''(y/x) - \frac{y}{x^3} \cdot f'''(y/x) \quad \dots(6)$$

$$\frac{\partial^2 z}{\partial y^3} = \frac{1}{x^2} f'''(y/x) \quad \dots(7)$$

Substitute (5) and (7) in (6), we get

$$\frac{\partial^2 z}{\partial y^2 \partial x} = -\frac{1}{x} \cdot \frac{\partial^2 z}{\partial y^2} - \frac{y}{x} \frac{\partial^2 z}{\partial y^3}$$

$$\text{i.e., } x \frac{\partial^2 z}{\partial y^2 \partial x} + \frac{\partial^2 z}{\partial y^2} + y \frac{\partial^2 z}{\partial y^3} = 0.$$

Exercise

Form partial differential equations by eliminating arbitrary constants.

1. $z = yf(x) + g(y)$

Ans. $p = yzs$.

2. $z = f(y) + g(x + y + z)$

Ans. $15r - 22s + 8t = 0$.

3. $z = f(x^2 + 3y) + g(x^2 - 3y)$

Ans. $9r - \frac{9}{x}p - 4x^2t = 0$.

4. $z = f(x + y) + g(x - y)$

Ans. $r = s$.

5. $xz = f(y - x) + g(x - y)$

1.3 Types of First Order Non-linear Partial Differential Equations

A PDE which involves the first order partial derivatives p and q with higher degree (> 1) and the products of p and q is called non-linear PDE. Some types of first order PDE is given below.

Type 1 : $F(p, q) = 0$.

Type 2 : $z = px + qy + f(p, q)$ [Clairaut's form]

Type 3 : $F(x, p, q) = 0$

Type 4 : $F(y, p, q) = 0$

Type 5 : $F(z, p, q) = 0$

Type 6 : $F_1(x, p) = F_2(y, q)$

Type 7 : $F(x^m p, y^n q) = 0$ and $F(z, x^m p, y^n q) = 0$

Type 8 : $F(z^m p, z^m q) = 0$ and $F_1(x, z^m p) = F_2(y, z^m q)$

Types of Solution:

A solution of a PDE is the relation between the independent and dependent variables (containing arbitrary constants or functions), which satisfies PDE. Also the solution of PDE is called as integral of PDE. According to that, the solutions are classified as follows.

A solution, which consists of maximum number of arbitrary functions is called a **general solution**.

A solution, which contains as many as arbitrary constants as the number of independent variables is called the **complete solution (or) complete integral**.

A solution is obtained by giving particular values to the arbitrary constants in the complete solution (or) to the arbitrary functions in the general solution is called **particular solution**.

Eliminate a between (4) and (5) gives the general integral.

Example 1.3.6. Solve : $p^2x^2 + q^2y^2 = z^2$

Apr.'15

Solution . Given equation is $p^2x^2 + q^2y^2 = z^2$.

...(1)

Since (1) can be reduced into the form $F(p, q) = 0$.

Dividing by z^2 on both sides we get

$$\begin{aligned}\frac{p^2x^2}{z^2} + \frac{q^2y^2}{z^2} &= 1 \\ \left(\frac{xp}{z}\right)^2 + \left(\frac{yq}{z}\right)^2 &= 1\end{aligned}\quad \dots(2)$$

Let $X = \log x$, $Y = \log y$ and $Z = \log z$.

$$\begin{aligned}\frac{\partial X}{\partial x} &= \frac{1}{x}; \quad \frac{\partial Y}{\partial y} = \frac{1}{y}; \quad \frac{\partial Z}{\partial z} = \frac{1}{z} \\ \frac{\partial X}{\partial x} &= \frac{\partial x}{x}; \quad \frac{\partial Y}{\partial y} = \frac{\partial y}{y}; \quad \frac{\partial Z}{\partial z} = \frac{\partial z}{z} \\ \frac{\partial Z}{\partial X} &= \left(\frac{\frac{\partial z}{z}}{\frac{\partial x}{x}}\right) = \frac{x}{z} \frac{\partial z}{\partial x} = \frac{xp}{z} \\ \frac{\partial Z}{\partial Y} &= \left(\frac{\frac{\partial z}{z}}{\frac{\partial y}{y}}\right) = \frac{y}{z} \frac{\partial z}{\partial y} = \frac{yq}{z}\end{aligned}$$

Substituting in (2) we get

$$\begin{aligned}\left(\frac{\partial Z}{\partial X}\right)^2 + \left(\frac{\partial Z}{\partial Y}\right)^2 &= 1 \\ \text{i.e., } P^2 + Q^2 &= 1.\end{aligned}$$

Now it is of the form $F(P, Q) = 1$.

Let the solution of (1) be $Z = aX + bY + c$.

...(2)

Partially differentiate (2) w.r.to X and Y we get $P = a$ and $Q = b$

Substitute $P = a$ and $Q = b$ in (2) we get

$$\begin{aligned}a^2 + b^2 &= 1. \text{ i.e., } b^2 = 1 - a^2. \\ \therefore b &= \pm \sqrt{1 - a^2}.\end{aligned}$$

Substitute b value in (2) we get

$$\begin{aligned} Z &= aX \pm \sqrt{1-a^2}Y + c \\ \log z &= a \log x \pm \sqrt{1-a^2} \log y + c \end{aligned} \quad \dots(3)$$

Which is the complete integral. Also there is no singular integral.

To find the General Integral :

In complete integral, put $c = \phi(a)$ then we get

$$\log z = a \log x \pm \sqrt{1-a^2} \log y + \phi(a) \quad \dots(4)$$

Partially differentiate (4) w.r.to a and equating to zero we get

$$\begin{aligned} \log x \pm \frac{-2a}{\sqrt{1-a^2}} \log y + \phi'(a) &= 0 \\ \log x \pm \frac{a \log y}{\sqrt{1-a^2}} + \phi'(a) &= 0. \end{aligned} \quad \dots(5)$$

Eliminate a between the equations (4) and (5) gives general integral.

Example 1.3.7. Solve : $x^4 p^2 - yzq = z^2$

...(1)

Solution . Given equation is $x^4 p^2 - yzq = z^2$.

Since (1) can be reduced into the form $F(p, q) = 0$.

Dividing by z^2 on both sides we get

$$\begin{aligned} \frac{x^4 p^2}{z^2} - \frac{yzq}{z^2} &= 1 \\ \left(\frac{x^2 p}{z}\right)^2 - \frac{yq}{z} &= 1 \end{aligned} \quad \dots(2)$$

Let $X = \frac{1}{x}$, $Y = \log y$ and $Z = \log z$.

$$\begin{aligned} \frac{\partial X}{\partial x} &= -\frac{1}{x^2}; \quad \frac{\partial Y}{\partial y} = \frac{1}{y}; \quad \frac{\partial Z}{\partial z} = \frac{1}{z} \\ \frac{\partial X}{\partial x} &= -\frac{\partial x}{x^2}; \quad \frac{\partial Y}{\partial y} = \frac{\partial y}{y}; \quad \frac{\partial Z}{\partial z} = \frac{\partial z}{z} \end{aligned}$$

$$\frac{\partial Z}{\partial X} = \left(\frac{\frac{\partial z}{\partial x}}{-\frac{\partial z}{\partial x^2}} \right) = -\frac{x^2}{z} \frac{\partial z}{\partial x} = -\frac{x^2 p}{z}$$

$$\therefore -\frac{\partial Z}{\partial X} = \frac{x^2 p}{z}$$

$$\frac{\partial Z}{\partial Y} = \left(\frac{\frac{\partial z}{\partial y}}{\frac{\partial z}{\partial y}} \right) = \frac{y}{z} \frac{\partial z}{\partial y} = \frac{y q}{z}$$

Substituting in (2) we get

$$\left(-\frac{\partial Z}{\partial X} \right)^2 - \left(\frac{\partial Z}{\partial Y} \right)^2 = 1$$

i.e., $P^2 - Q^2 = 1$.

Now it is of the form $F(P, Q) = 1$.

Let the solution of (1) be $Z = aX + bY + c$(2)

Partially differentiate (2) w.r.to X and Y we get $P = a$ and $Q = b$
Substitute $P = a$ and $Q = b$ in (2) we get

$$a^2 - b^2 = 1. \therefore b^2 = a^2 - 1.$$

Substitute b value in (2) we get

$$Z = aX + (a^2 - 1)Y + c$$

$$\log z = a \log x + (a^2 - 1) \log y + c \quad \text{...(3)}$$

Which is the complete integral.

Exercise

Solve the following equations.

1. $p^2 + p = q^2$

Ans. $z = ax + \sqrt{a^2 + a}y + c$

2. $p^2 + q^2 = 9$

Ans. $z = ax + \sqrt{9 - a^2}y + c$

3. $pq + p + q = 0$

Ans. $z = ax - \frac{a}{a+1}y + c$

4. $p = 1 + q^2$

Ans. $z = ax + \sqrt{a-1}y + c$

5. $p = q^2$

Ans. $z = ax + \sqrt[3]{a}y + c$

1.3.2 Type 2 : Equation is of the Clairaut's Form

$$z = px + qy + f(p, q)$$

Working rule : Check the given equation is Clairaut's form

$$z = px + qy + f(p, q).$$

...(1)

To find Complete Integral :

Replace $p = a$ and $q = b$ in (1) we get $z = ax + by + f(a, b).$

...(2)

Where a and b are arbitrary constants.

To find Singular Integral :

Partially differentiate (2) w.r.to a, b and equating to 0, we get

$$\frac{\partial z}{\partial a} = x + f'(a, b) = 0$$

...(3)

$$\frac{\partial z}{\partial b} = y + f'(a, b) = 0$$

...(4)

Eliminate a and b from the equations (2), (3) and (4), then we get singular integral.

To find the General Integral :

In complete integral, put $b = \phi(a)$ then we get

$$z = ax + \phi(a)y + f(a, \phi(a)).$$

...(5)

Partially differentiate (5) w.r.to a and equating to zero we get

$$z = ax + \phi'(a)y + f'(a, \phi(a)).$$

...(6)

Eliminate a between the equations (5) and (6) gives general integral

Example 1.3.8. Solve : $z = px + qy + p^2 - q^2.$

(Nov.'12)

Solution . Given equation is $z = px + qy + p^2 - q^2.$

...(1)

This is of the Clairaut's form $z = px + qy + f(p, q)$

$\therefore$ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + a^2 - b^2$$

...(2)

$$\frac{\partial z}{\partial a} = x + b = 0. \Rightarrow b = -x \quad \text{and} \quad \frac{\partial z}{\partial b} = y + a = 0. \Rightarrow a = -y.$$

Substitute a and b values in (2), we get

$$z = -xy - xy + xy$$

$$\therefore z = -xy. \quad \text{It is a singular solution.}$$

To find the General Integral :

In complete integral, put $b = \phi(a)$ then we get

$$z = ax + \phi(a)y + a\phi(a). \quad \dots(3)$$

Partially differentiate (3) w.r.to a and equating to zero we get

$$z = x + \phi'(a)y + \phi(a) + a\phi'(a). \quad \dots(4)$$

Eliminate a between the equations (3) and (4) gives general integral

Example 1.3.11. Find complete and singular integral of

$$z = px + qy + p^2 + pq + q^2. \quad (\text{Nov.'11})$$

Solution . Given equation is $z = px + qy + p^2 + pq + q^2. \quad \dots(1)$

This is of the Clairaut's form $z = px + qy + f(p, q)$

$\therefore$ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + a^2 + ab + b^2 \quad \dots(2)$$

To find Singular Integral :

Partially differentiate (2) w.r.to a, b and equating to 0, we get

$$\frac{\partial z}{\partial a} = x + 2a + b = 0.$$

$$\frac{\partial z}{\partial b} = y + a + 2b = 0.$$

Solving the above two equation then we get

$$a = \frac{y - 2x}{3} \quad \text{and} \quad b = \frac{x - 2y}{3}$$

Substitute a and b values in (2), we get

$$\begin{aligned}
 y - 2xz &= \left(\frac{y-2x}{3}\right)x + \left(\frac{x-2y}{3}\right)y + \left(\frac{y-2x}{3}\right)^2 \\
 &\quad + \left(\frac{y-2x}{3}\right)\left(\frac{x-2y}{3}\right) + \left(\frac{x-2y}{3}\right)^2 \\
 &= \frac{yx - 2x^2}{3} + \frac{xy - 2y^2}{3} + \frac{y^2 - 4xy + 4x^2}{9} \\
 &\quad + \frac{xy - 2y^2 - 2x^2 + 4xy}{9} + \frac{x^2 - 4xy + 4y^2}{3} \\
 &= \left\{ \frac{3xy - 6x^2 + 3xy - 6y^2 + y^2 + 4x^2 - 4xy}{9} \right. \\
 &\quad \left. + \frac{xy - 2y^2 - 2x^2 + 4xy + x^2 + 4y^2 - 4xy}{9} \right\} \\
 &= \frac{3xy - 3x^2 - 3y^2}{9} \\
 \therefore 3z &= xy - x^2 - y^2.
 \end{aligned}$$

It is a singular solution.

Example 1.3.12. Find complete and singular integral of
 $z = px + qy + \left(\frac{q}{p} - p\right).$ (Apr.'14)

Solution . Given equation is $z = px + qy + \left(\frac{q}{p} - p\right).$... (1)

This is of the Clairaut's form $z = px + qy + f(p, q)$

$\therefore$ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + \left(\frac{b}{a} - a\right) \quad \dots (2)$$

To find Singular Integral :

Partially differentiate (2) w.r.to a, b and equating to 0, we get

$$\begin{aligned}
 \frac{\partial z}{\partial a} &= x - \frac{b}{a^2} - 1 = 0. \quad \dots (3) \\
 \frac{\partial z}{\partial b} &= y + \frac{1}{a} = 0. \Rightarrow a = -\frac{1}{y}
 \end{aligned}$$

Substitute $a = -\frac{1}{y}$ in (3) we get

$$x - \frac{b}{\frac{1}{y^2}} - 1 = 0$$

$$x - by^2 - 1 = 0$$

$$\Rightarrow b = \frac{x-1}{y^2}.$$

Substitute a and b values in (2) we get

$$\begin{aligned} z &= \left(-\frac{1}{y}\right)x + \left(\frac{x-1}{y^2}\right)y + \left(\frac{\frac{x-1}{y^2}}{-\frac{1}{y}} + \frac{1}{y}\right) \\ &= -\frac{x}{y} + \frac{x-1}{y} + \left(\frac{-(x-1)}{y} + \frac{1}{y}\right) \\ &= \frac{1-x}{y} \end{aligned}$$

$$\therefore yz = 1-x.$$

It is a singular solution.

Example 1.3.13. Solve : $z = px + qy + 2\sqrt{pq}$.

Solution . Given equation is $z = px + qy + 2\sqrt{pq}$ (1)

This is of the Clairaut's form $z = px + qy + f(p, q)$

$\therefore$ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + 2\sqrt{ab}. \quad \dots (2)$$

To find Singular Integral :

Partially differentiate (2) w.r.to a, b and equating to 0, we get

$$\frac{\partial z}{\partial a} = x + 2 \frac{1}{\sqrt{ab}} \times b = 0.$$

$$\text{i.e., } x + \sqrt{\frac{b}{a}} = 0. \Rightarrow x = -\sqrt{\frac{b}{a}}. \quad \dots (3)$$

$$\frac{\partial z}{\partial b} = y + 2 \frac{1}{\sqrt{ab}} \times a = 0.$$

$$\text{i.e., } y + \sqrt{\frac{a}{b}} = 0. \Rightarrow y = -\sqrt{\frac{a}{b}}. \quad \dots (4)$$

1.3.5 Type 5 : $F(z, p, q) = 0$

In this type the variables x and y do not appear explicitly.

Working rule :

1. Check the given equation is of the form $F(z, p, q) = 0$... (1)

2. Let $z = f(x + ay)$ be the solution of (1).

Put $u = ax + y$ then $z = f(u)$... (2)

3. Differentiate (2) partially w.r.to x and y we get

$$\begin{aligned} p &= \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right) \\ q &= \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right) \\ \therefore p &= \frac{dz}{du}, \quad q = a \frac{dz}{du} \end{aligned} \quad \dots (3)$$

4. Substituting (3) in (1) we get an ordinary differential equation is of the form $f\left(z, \frac{dz}{du}, a \frac{dz}{du}\right) = 0$... (4)

5. Solve (4), by using variable separable method we have

$\phi(z, a) = x + ay + b$. Which is a complete integral.

6. Find the singular and general integral as usual way.

Method : 2

Working rule :

1. Check the given equation is of the form $F(z, p, q) = 0$... (1)

2. Replace $q = ap$ and rewrite (1) as p in terms of z and a .
i.e., $p = \phi(z, a)$

3. Now, consider $dz = p dx + q dy$ then replace $q = ap$ we get
 $dz = p dx + ap dy$. That is, $dz = p(dx + a dy)$... (2)

4. Substitute $p = \phi(z, a)$ in (2) and applying variable separable method then integrate we have

$$\int \frac{dz}{\phi(z, a)} = \int (dx + a dy) + b$$

$$\int \frac{dz}{\phi(z, a)} = (x + ay) + b$$

Which is a complete integral.

5. Find the singular and general integral as usual way.

Example 1.3.24. Solve : $p(1 + q) = qz$

Apr.'11

Solution . Given equation is $p(1 + q) = qz$.

...(1)

This is of the type $f(z, p, q) = 0$.

Put $u = x + ay$ then $z = f(u)$

...(2)

Differentiate (2) partially w.r.to x and y we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right)$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right)$$

$$\therefore p = \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \text{...(3)}$$

Substituting (3) in (1) we get

$$\frac{dz}{du} \left(1 + a \frac{dz}{du} \right) = a \frac{dz}{du} z$$

$$\left(1 + a \frac{dz}{du} \right) = a z$$

$$\frac{dz}{du} = \frac{az - 1}{a} = z - \frac{1}{a}$$

$$\therefore \frac{dz}{z - \frac{1}{a}} = du$$

$$\begin{aligned}\text{Integrating, } \int \frac{dz}{z - \frac{1}{a}} &= \int du + c \\ \log \left(z - \frac{1}{a} \right) &= u + c \\ \log \left(z - \frac{1}{a} \right) &= x + ay + c \quad [\because u = x + ay]\end{aligned}$$

Which is a complete integral.

Method 2 :

$$\text{Given equation is } p(1 + q) = qz. \quad \dots(1)$$

This is of the type $f(z, p, q) = 0$.

Replace $q = ap$ in (1) we get

$$\begin{aligned}p(1 + q) &= qz \\ p(1 + ap) &= apz \\ (1 + ap) &= az \\ \therefore ap &= az - 1 \\ \Rightarrow p &= \frac{az - 1}{a} = z - \frac{1}{a} \quad \dots(2)\end{aligned}$$

Consider $dz = p dx + q dy$. Substitute $q = ap$ we get

$$\begin{aligned}dz &= p dx + ap dy = p(dx + a dy) \\ \therefore dz &= \left(z - \frac{1}{a} \right) [dx + a dy] \\ \frac{dz}{z - \frac{1}{a}} &= dx + a dy \\ \text{Integrating, } \int \frac{dz}{z - \frac{1}{a}} &= \int (dx + a dy) + c \\ \log \left(z - \frac{1}{a} \right) &= (x + ay) + c.\end{aligned}$$

Which is the complete integral.

Example 1.3.25. Solve : $9(p^2 z + q^2) = 4$
May'00, Nov.'09, Nov.'10

Consider $dz = p dx + q dy$. Substitute $q = ap$ we get

$$dz = p dx + ap dy = p(dx + a dy)$$

$$\therefore dz = \pm \frac{2}{3\sqrt{z+a^2}} [dx + a dy]$$

$$\sqrt{z+a^2} dz = \pm \frac{2}{3} [dx + a dy]$$

Integrating, $\int \sqrt{z+a^2} dz = \pm \frac{2}{3} \int (dx + a dy) + b$

$$\frac{(z+a^2)^{3/2}}{3/2} = \pm \frac{2}{3} (x + ay) + b$$

$$\frac{2}{3} (z+a^2)^{3/2} = \pm \frac{2}{3} (x + ay) + b$$

$$(z+a^2)^{3/2} = \pm (x + ay) + c \quad [\because b = 2c/3]$$

Which is the complete integral.

Example 1.3.26. Solve : $p^2 + pq = z^2$

Nov.'14

Solution . Given equation is $p^2 + pq = z^2$ (1)

This is of the type $f(z, p, q) = 0$.

Put $u = x + ay$ then $z = f(u)$... (2)

Differentiate (2) partially w.r.to x and y we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right)$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right)$$

$$\therefore p = \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \dots (3)$$

Substituting (3) in (1) we get

$$\left(\frac{dz}{du} \right)^2 + \frac{dz}{du} \left(a \frac{dz}{du} \right) = z^2$$

$$\left(\frac{dz}{du} \right)^2 (1 + a) = z^2$$

$$\left(\frac{dz}{du} \right)^2 = \frac{z^2}{1+a}$$

$$\begin{aligned}\frac{dz}{du} &= \frac{z}{\sqrt{1+a}} \\ \frac{dz}{z} &= \frac{du}{\sqrt{1+a}} \\ \text{Integrating, } \int \frac{dz}{z} &= \frac{1}{\sqrt{1+a}} \int du + c \\ \log z &= \frac{u}{\sqrt{1+a}} + c \\ \log z &= \frac{x+ay}{\sqrt{1+a}} + c \quad [\because u = x+ay]\end{aligned}$$

Which is a complete integral.

Example 1.3.27. Solve : $z^2 = p^2 + q^2 + 1$

Solution . Given equation is $z^2 = p^2 + q^2 + 1$ (1)

This is of the type $f(z, p, q) = 0$ (2)

Put $u = x + ay$ then $z = f(u)$

Differentiate (2) partially w.r.to x and y we get

$$\begin{aligned}p &= \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right) \\ q &= \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right) \\ \therefore p &= \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \dots (3)\end{aligned}$$

Substituting (3) in (1) we get

$$\begin{aligned}z^2 &= \left(\frac{dz}{du} \right)^2 + \left(a \frac{dz}{du} \right)^2 + 1 \\ \left(\frac{dz}{du} \right)^2 (1 + a^2) &= z^2 - 1 \\ \left(\frac{dz}{du} \right)^2 &= \frac{z^2 - 1}{1 + a^2} \\ \frac{dz}{du} &= \sqrt{\frac{z^2 - 1}{1 + a^2}} \\ \frac{dz}{\sqrt{z^2 - 1}} &= \frac{1}{\sqrt{1 + a^2}} du\end{aligned}$$

$$\begin{aligned}\text{Integrating, } \int \frac{dz}{\sqrt{z^2 - 1}} &= \frac{1}{\sqrt{1 + a^2}} \int du + c \\ \cosh^{-1} z &= \frac{1}{\sqrt{1 + a^2}} u + c \\ \cosh^{-1} z &= \frac{1}{\sqrt{1 + a^2}} (x + ay) + c \quad [\because u = x + ay]\end{aligned}$$

Which is a complete integral.

Example 1.3.28. Solve : $p^3 + q^3 = 8z$

Solution . Given equation is $p^3 + q^3 = 8z$(1)

This is of the type $f(z, p, q) = 0$.

Put $u = x + ay$ then $z = f(u)$...(2)

Differentiate (2) partially w.r.to x and y we get

$$\begin{aligned}p &= \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right) \\ q &= \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right) \\ \therefore p &= \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \text{...(3)}\end{aligned}$$

Substituting (3) in (1) we get

$$\left(\frac{dz}{du} \right)^3 + \left(a \frac{dz}{du} \right)^3 = 8z$$

$$\left(\frac{dz}{du} \right)^3 (1 + a^3) = 8z$$

$$\left(\frac{dz}{du} \right)^3 = \frac{8z}{1 + a^3}$$

$$\frac{dz}{du} = \frac{2z^{1/3}}{(1 + a^3)^{1/3}}$$

$$\frac{dz}{z^{1/3}} = \frac{2}{(1 + a^3)^{1/3}} du$$

$$\text{Integrating, } \int z^{-1/3} dz = \frac{2}{(1 + a^3)^{1/3}} \int du + c$$

$$\left(\frac{dz}{du}\right)(a + bk) = -cz$$

$$\frac{dz}{z} = -\frac{c}{(a + bk)} du$$

Integrating, $\int \frac{dz}{z} = -\frac{c}{(a + bk)} \int du + \log c_1$

$$\log z = -Du + \log c_1 \quad \left[\because D = \frac{c}{(a + bk)} \right]$$

$$\log \left(\frac{z}{c_1} \right) = -Du$$

$$\frac{z}{c_1} = e^{-Du}$$

$$z = c_1 e^{-D(x+ky)} \quad [\because u = x + ky]$$

Which is a complete integral.

Example 1.3.34. Solve : $p(1 + q^2) = q(z - a)$

Apr.'15

Solution . Given equation is $p(1 + q^2) = q(z - a)$ (1)

This is of the type $f(z, p, q) = 0$.

Put $u = x + ay$ then $z = f(u)$... (2)

Differentiate (2) partially w.r.to x and y we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right)$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right)$$

$$\therefore p = \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \dots (3)$$

Substituting (3) in (1) we get

$$\left(\frac{dz}{du}\right) \left[1 + \left(a \frac{dz}{du}\right)^2 \right] = \left(a \frac{dz}{du}\right) (z - a)$$

$$1 + \left(a \frac{dz}{du}\right)^2 = a(z - a)$$

$$\left(a \frac{dz}{du}\right)^2 = a(z - a) - 1 = az - a^2 - 1$$

$$a \frac{dz}{du} = \sqrt{az - (a^2 + 1)}$$

$$a \frac{dz}{du} = \sqrt{a} \sqrt{z - \left(\frac{a^2 + 1}{a}\right)}$$

$$\frac{dz}{du} = \frac{1}{\sqrt{a}} \sqrt{z - k}$$

$$\left[\because k = \frac{a^2 + 1}{a} \right]$$

$$\frac{dz}{\sqrt{z - k}} = \frac{1}{\sqrt{a}} du$$

Integrating, $\int \frac{dz}{\sqrt{z - k}} = \frac{1}{\sqrt{a}} \int du + c$

$$\int (z - k)^{-1/2} dz = \frac{1}{\sqrt{a}} u + c$$

$$2(z - k)^{1/2} = \frac{1}{\sqrt{a}} u + c$$

$$2 \sqrt{z - \left(\frac{a^2 + 1}{a}\right)} = \frac{x + ay}{\sqrt{a}} + c \quad [\because u = x + ay]$$

Which is a complete integral.

Example 1.3.35. Solve : $pz = 1 + q^2$

Solution . Given equation is $pz = 1 + q^2$(1)

This is of the type $f(z, p, q) = 0$.

Put $u = x + ay$ then $z = f(u)$...(2)

Differentiate (2) partially w.r.to x and y we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 \quad \left(\because \frac{\partial u}{\partial x} = 1 \right)$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a \quad \left(\because \frac{\partial u}{\partial y} = a \right)$$

$$\therefore p = \frac{dz}{du}, \quad q = a \frac{dz}{du} \quad \text{...(3)}$$

Substituting (3) in (1) we get

$$\left(\frac{dz}{du} \right) z = 1 + \left(a \frac{dz}{du} \right)^2$$

$$a^2 \left(\frac{dz}{du} \right)^2 - z \left(\frac{dz}{du} \right) + 1 = 0$$

$$\frac{dz}{du} = \frac{z \pm \sqrt{z^2 - 4a^2}}{2a^2}$$

$$\frac{dz}{z + \sqrt{z^2 - 4a^2}} = \frac{du}{2a^2}$$

$$\frac{(z - \sqrt{z^2 - 4a^2}) dz}{(z + \sqrt{z^2 - 4a^2})(z - \sqrt{z^2 - 4a^2})} = \frac{du}{2a^2}$$

$$\frac{(z - \sqrt{z^2 - 4a^2}) dz}{z^2 - (z^2 - 4a^2)} = \frac{du}{2a^2}$$

$$\frac{(z - \sqrt{z^2 - 4a^2}) dz}{4a^2} = \frac{du}{2a^2}$$

$$(z - \sqrt{z^2 - 4a^2}) dz = 2 du$$

Integrating, $\int (z - \sqrt{z^2 - 4a^2}) dz = 2 \int du + c$

$$\frac{z^2}{2} - \int \sqrt{z^2 - 4a^2} dz = 2u + c$$

$$\frac{z^2}{2} - \left[\frac{z\sqrt{z^2 - 4a^2}}{2} - \frac{4a^2}{2} \cosh^{-1} \left(\frac{z}{2a} \right) \right] = 2(x + ay) + c$$

Which is a complete integral.

Exercise

Solve the following equations.

1. $z = pq$

Ans. $4az = (x + ay + c)^2$.

2. $z = p^2 + q^2$

Ans. $4z = \frac{1}{1+a^2}(x + ay + c)$.

3. $p(1 - q^2) = q(1 - z)$

Ans. $4(1 - a + az) = (x + ay + c)$.

4. $p^3 + q^3 = z^3$

Ans. $\log(az - 1) = (x + ay + c)$.

5. $p^3 + q^3 = 3zpq$

Ans. $(1 + a^3) \log z = 3a(x + ay) + c$.

6. $p^2 + qz = z^2$

Ans. $\log z = \frac{-a \pm \sqrt{a^2 + 4}}{2} (x + ay) + c$.

Example 1.3.48. Solve : $x^2 p^2 + y^2 q^2 = z^2$.

Apr.'15

Solution . Given equation is $x^2 p^2 + y^2 q^2 = z^2$.

$$\text{i.e., } (xp)^2 + (yq)^2 = z^2. \quad \dots(1)$$

This is of the type $F(z, x^m p, y^n q) = 0$. Where $m = 1, n = 1$

$$\text{Put } X = \log x$$

$$\frac{\partial X}{\partial x} = \frac{1}{x}$$

$$\text{Let } P = \frac{\partial z}{\partial X}$$

$$\text{Then } \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \frac{\partial X}{\partial x}$$

$$\therefore p = P \cdot \frac{1}{x}$$

$$\Rightarrow P = xp \quad \dots(2)$$

$$\text{Put } Y = \log y$$

$$\frac{\partial Y}{\partial y} = \frac{1}{y}$$

$$\text{Let } Q = \frac{\partial z}{\partial Y}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial Y} \frac{\partial Y}{\partial y}$$

$$q = Q \cdot \frac{1}{y}$$

$$Q = yq \quad \dots(3)$$

Substituting (2) and (3) in (1) we get $P^2 + Q^2 = z^2$(4)

Now this is of the type $F(z, P, Q) = 0$.

Replace $Q = aP$ in (4) we get

$$P^2 + a^2 P^2 = z^2.$$

$$\text{i.e., } P^2(1 + a^2) = z^2$$

$$\therefore P^2 = \frac{z^2}{(1 + a^2)}$$

$$\Rightarrow P = \pm \frac{z}{\sqrt{1 + a^2}}$$

Consider $dz = PdX + QdY$. Substitute $Q = aP$ we get

$$dz = PdX + aPdY = P(dX + adY)$$

$$\therefore dz = \pm \frac{z}{\sqrt{1 + a^2}} (dX + adY)$$

$$\frac{dz}{z} = \pm \frac{1}{\sqrt{1 + a^2}} (dX + adY)$$

$$\text{Integrating, } \int \frac{dz}{z} = \pm \frac{1}{\sqrt{1 + a^2}} \int (dX + adY) + c$$

$$\log z = \pm \frac{1}{\sqrt{1+a^2}}(X + aY) + c$$

$$\log z = \pm \frac{1}{\sqrt{1+a^2}}(\log x + a \log y) + c$$

$$[\because X = \log x; Y = \log y]$$

Which is the complete integral

Example 1.3.49. Solve : $q^2 y^2 = z(z - px)$.

Solution . Given equation is $q^2 y^2 = z(z - px)$.

$$\text{i.e., } zpx + q^2 y^2 = z^2.$$

This is of the type $F(z, x^m p, y^n q) = 0$. Where $m = 1, n = 1$

$$\text{Put } X = \log x$$

$$\frac{\partial X}{\partial x} = \frac{1}{x}$$

$$\text{Let } P = \frac{\partial z}{\partial X}$$

$$\text{Then } \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \frac{\partial X}{\partial x}$$

$$\therefore p = P \cdot \frac{1}{x}$$

$$\Rightarrow P = xp \quad \dots(2)$$

$$\text{Put } Y = \log y$$

$$\frac{\partial Y}{\partial y} = \frac{1}{y}$$

$$\text{Let } Q = \frac{\partial z}{\partial Y}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial Y} \frac{\partial Y}{\partial y}$$

$$q = Q \cdot \frac{1}{y}$$

$$Q = yq \quad \dots(3)$$

Substituting (2) and (3) in (1) we get $zP + Q^2 = z^2$.

Now this is of the type $F(z, P, Q) = 0$.

Replace $Q = aP$ in (4) we get

$$zP + a^2 P^2 = z^2.$$

$$\text{i.e., } a^2 P^2 + zP - z^2 = 0$$

$$P = \frac{-z \pm \sqrt{z^2 + 4a^2 z^2}}{2a^2}$$

$$\therefore P = z \left(\frac{-1 \pm \sqrt{1 + 4a^2}}{2a^2} \right)$$

From (3) and (4), the general solution is $\phi(3x - 2y, y - 3z) = 0$.

Example 1.3.63. Solve : $xp + yq = z$.

(Nov.'10)

Solution . Given equation is $xp + yq = z$.

...(1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\begin{aligned} \frac{dx}{P} &= \frac{dy}{Q} = \frac{dz}{R} \\ \frac{dx}{x} &= \frac{dy}{y} = \frac{dz}{z} \end{aligned} \quad \dots(2)$$

Where $P = x$, $Q = y$ and $R = z$.

Taking first and second ratio, we have

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating we get

$$\begin{aligned} \int \frac{dx}{x} &= \int \frac{dy}{y} + \log c_1 \\ \log x &= \log y + \log c_1 \\ \log \frac{x}{y} &= \log c_1 \\ \Rightarrow \frac{x}{y} &= c_1. \end{aligned} \quad \dots(3)$$

Taking second and third ratio, we have

$$\frac{dy}{y} = \frac{dz}{z}$$

Integrating we get

$$\begin{aligned} \int \frac{dy}{y} &= \int \frac{dz}{z} + \log c_2 \\ \log y &= \log z + \log c_2 \\ \log \frac{y}{z} &= \log c_2 \\ \Rightarrow \frac{y}{z} &= c_2. \end{aligned} \quad \dots(4)$$

From (3) and (4), the general solution is $\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$.

$$\frac{xy}{2} = \frac{z^2}{2} + c_2 \Rightarrow xy - z^2 = 2c_2 \quad \dots(4)$$

From (3) and (4), the general solution is $\phi\left(\frac{x}{y}, xy - z^2\right) = 0$.

Example 1.3.66. Solve : $\frac{y^2 z}{x} p + xzq = y^2$.

(Nov. '01)

Solution . Given equation is $\frac{y^2 z}{x} p + xzq = y^2$.

$\dots(1)$

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \dots(2)$$

Where $P = \frac{y^2 z}{x}$, $Q = xz$ and $R = y^2$.

Taking first and second ratio, we have

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz}$$

$$\frac{x dx}{y^2} = \frac{dy}{x}$$

$$x^2 dx = y^2 dy$$

Integrating, $\int x^2 dx = \int y^2 dy + c$

$$\frac{x^3}{3} = \frac{y^3}{3} + c$$

$$x^3 - y^3 = c_1 \quad [\because c_1 = 3c] \quad \dots(3)$$

Taking first and third ratio, we have

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dz}{y^2}$$

$$\frac{x dx}{y^2 z} = \frac{dz}{y^2}$$

$$x dx = z dz$$

Integrating, $\int x dx = \int z dz + c$

$$\frac{x^2}{2} = \frac{z^2}{2} + c_2$$

$$x^2 - z^2 = c_3 \quad [\because c_3 = 2c_2] \quad \dots(4)$$

From (3) and (4), the general solution is $\phi(x^3 - y^3, x^2 - y^2) = 0$.

Example 1.3.67. Solve : $(1 - x)p + (2 - y)q = 3 - z$.

Solution . Given equation is $(1 - x)p + (2 - y)q = 3 - z$ (1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{1 - x} = \frac{dy}{2 - y} = \frac{dz}{3 - z} \quad \dots (2)$$

Where $P = 1 - x$, $Q = 2 - y$ and $R = 3 - z$.

Taking first and second ratio, we have

$$\frac{dx}{1 - x} = \frac{dy}{2 - y}$$

Integrating, $\int \frac{dx}{1 - x} = \int \frac{dy}{2 - y} + \log c_1$

$$-\log(1 - x) = -\log(2 - y) + \log c_1$$

$$\log \frac{2 - y}{1 - x} = \log c_1$$

$$\Rightarrow \frac{2 - y}{1 - x} = c_1. \quad \dots (3)$$

Taking second and third ratio, we have

$$\frac{dy}{2 - y} = \frac{dz}{3 - z}$$

Integrating, $\int \frac{dy}{2 - y} = \int \frac{dz}{3 - z} + \log c_2$

$$-\log(2 - y) = -\log(3 - z) + \log c_2$$

$$\log \frac{3 - z}{2 - y} = \log c_2$$

$$\Rightarrow \frac{3 - z}{2 - y} = c_2. \quad \dots (4)$$

From (3) and (4), the general solution is $\phi\left(\frac{2 - y}{1 - x}, \frac{3 - z}{2 - y}\right) = 0$.

(Apr.'06)

Example 1.3.68. Solve : $\cot x p + \cot y q = \cot z$.

Problems Based on Method of Multipliers :

Example 1.3.69. Solve : $(3z - 4y)p + (4x - 2z)q = 2y - 3x$. (or)
 $(3z - 4y)\frac{\partial z}{\partial x} + (4x - 2z)\frac{\partial z}{\partial y} = 2y - 3x$.

(Apr.'09, Nov.'11, Apr.'12, Nov.'14)

Solution . Given equation $(3z - 4y)p + (4x - 2z)q = 2y - 3x$... (1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{3z - 4y} = \frac{dy}{4x - 2z} = \frac{dz}{2y - 3x} \quad \dots (2)$$

Where $P = 3z - 4y$, $Q = 4x - 2z$ and $R = 2y - 3x$.

Choose a set of multipliers x , y and z . Then

$$\begin{aligned} \text{Each of (2)} &= \frac{xdx + ydy + zdz}{x(3z - 4y) + y(4x - 2z) + z(2y - 3x)} \\ &= \frac{xdx + ydy + zdz}{3xz - 4xy + 4yz - 2yz + 2yz - 3xz} \\ &= \frac{xdx + ydy + zdz}{0} \end{aligned}$$

$$xdx + ydy + zdz = 0$$

Integrating we get,

$$\begin{aligned} \int xdx + \int ydy + \int zdz &= c \\ \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} &= c \\ x^2 + y^2 + z^2 &= c_1 \quad [\because c_1 = 2c] \end{aligned} \quad \dots (3)$$

Similarly choose another set of multipliers 2, 3 and 4. Then each ratio of (2) are equal to

$$\begin{aligned} \frac{2dx + 3dy + 4dz}{2(3z - 4y) + 3(4x - 2z) + 4(2y - 3x)} &= \frac{2dx + 3dy + 4dz}{\begin{Bmatrix} 6z - 8y + 12z \\ -6z + 8y - 12z \end{Bmatrix}} \\ &= \frac{2dx + 3dy + 4dz}{0} \end{aligned}$$

$$\therefore 2dx + 3dy + 4dz = 0$$

$$\text{Integrating, } 2 \int dx + 3 \int dy + 4 \int dz = c_2$$

$$2x + 3y + 4z = c_2. \quad \dots(4)$$

From the equations (3) and (4), the general solution is $\phi(x^2 + y^2 + z^2, 2x + 3y + 4z) = 0$.

Example 1.3.70. Solve : $(y - z)p + (z - x)q = (x - y)$.
(Nov.'03, Apr.'15)

Solution . Given equation is $(y - z)p + (z - x)q = (x - y)$. $\dots(1)$

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{y - z} = \frac{dy}{z - x} = \frac{dz}{x - y} \quad \dots(2)$$

Where $P = y - z$, $Q = z - x$ and $R = x - y$.

Choose multipliers 1, 1, 1. From each ratio of (2) is equal to

$$\frac{dx + dy + dz}{y - z + z - x + x - y} = \frac{dx + dy + dz}{0}$$

$$d(x + y + z) = 0$$

$$\text{Integrating, } \int d(x + y + z) = c_1 \quad \dots(3)$$

$$x + y + z = c_1.$$

Similarly choose another set of multipliers x , y and z . Then each ratio of (2) are equal to

$$\begin{aligned} \frac{xdx + ydy + zdz}{x(y - z) + y(z - x) + z(x - y)} &= \frac{xdx + ydy + zdz}{xy - xz + yz - yx + zx - zy} \\ &= \frac{xdx + ydy + zdz}{0} \end{aligned}$$

$$xdx + ydy + zdz = 0$$

$$\int xdx + \int ydy + \int zdz = c$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 + y^2 + z^2 = c_2$$

$$[c_2 = 2c]$$

$\dots(4)$

From the equations (3) and (4), the general solution is $\phi(x + y + z, x^2 + y^2 + z^2) = 0$.

Example 1.3.71. Solve : $x(y - z)p + y(z - x)q = z(x - y)$. (Nov.'06)

Solution . Given equation is $x(y - z)p + y(z - x)q = z(x - y)$...(1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x(y - z)} = \frac{dy}{y(z - x)} = \frac{dz}{z(x - y)} \quad \dots(2)$$

Where $P = x(y - z)$, $Q = y(z - x)$ and $R = z(x - y)$.

Choose a set of multipliers 1, 1 and 1. Then each ratio of (2) are equal to

$$\frac{dx + dy + dz}{x(y - z) + y(z - x) + z(x - y)} = \frac{dx + dy + dz}{xy - xz + yz - xy + xz - yz}$$

$$= \frac{dx + dy + dz}{0}$$

$$d(x + y + z) = 0$$

$$\text{Integrating, } \int d(x + y + z) = c_1$$

$$x + y + z = c_1. \quad \dots(3)$$

Similarly choose another set of multipliers $\frac{1}{x}$, $\frac{1}{y}$ and $\frac{1}{z}$. Then each ratio of (2) are equal to

$$\frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{(y - z) + (z - x) + (x - y)} = \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}$$

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

$$\text{Integrating, } \int \frac{dx}{x} + \int \frac{dy}{y} + \int \frac{dz}{z} = \log c_2$$

$$\log x + \log y + \log z = \log c_2$$

$$\log xyz = \log c_2 \quad \dots(4)$$

$$\therefore xyz = c_2.$$

From (3) and (4), the general solution is $\phi(x + y + z, xyz) = 0$.

Example 1.3.72. Solve : $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.
(or) $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$.

(May '00, Apr.'03, Apr.'05, Nov.'09, Apr.'10, Apr.'11, Nov.'12, Apr.'15)

Solution . Given equation is $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.
This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)} \quad \dots(1)$$

Where $P = x(y^2 - z^2)$, $Q = y(z^2 - x^2)$ and $R = z(x^2 - y^2)$.

Choose a set of multipliers x , y and z . Then each ratio of (1) are equal to

$$\begin{aligned} \frac{xdx + ydy + zdz}{x^2(y^2 - z^2) + y^2(z^2 - x^2) + z^2(x^2 - y^2)} &= \frac{xdx + ydy + zdz}{\begin{bmatrix} x^2y^2 - x^2z^2 + y^2z^2 \\ -x^2y^2 + x^2z^2 - y^2z^2 \end{bmatrix}} \\ &= \frac{xdx + ydy + zdz}{0} \end{aligned}$$

$$xdx + ydy + zdz = 0$$

Integrating we get,

$$\int xdx + \int ydy + \int zdz = c$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 + y^2 + z^2 = c_1 \quad [\because c_1 = 2c] \quad \dots(2)$$

Similarly choose another set of multipliers $\frac{1}{x}$, $\frac{1}{y}$ and $\frac{1}{z}$. Then each ratio of (1) are equal to

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{(y^2 - z^2) + (z^2 - x^2) + (x^2 - y^2)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

$$\begin{aligned}
 \text{Integrating, } \int \frac{dx}{x} + \int \frac{dy}{y} + \int \frac{dz}{z} &= \log c_2 \\
 \log x + \log y + \log z &= \log c_2 \\
 \log xyz &= \log c_2 \\
 \therefore xyz &= c_2. \quad \dots(3)
 \end{aligned}$$

From (2) and (3), the general solution is $\phi(x^2 + y^2 + z^2, xyz) = 0$.

Example 1.3.73. Solve : $\left(\frac{y-z}{yz}\right)p + \left(\frac{z-x}{zx}\right)q = \left(\frac{x-y}{xy}\right)$.

(Apr.'07, Nov.'13)

Solution . Given equation is $\left(\frac{y-z}{yz}\right)p + \left(\frac{z-x}{zx}\right)q = \left(\frac{x-y}{xy}\right)$.

Multiplying both sides by xyz then we get

$$x(y-z)p + y(z-x)q = z(x-y). \quad \dots(1)$$

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\begin{aligned}
 \frac{dx}{P} &= \frac{dy}{Q} = \frac{dz}{R} \\
 \frac{dx}{x(y-z)} &= \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} \quad \dots(2)
 \end{aligned}$$

Where $P = x(y-z)$, $Q = y(z-x)$ and $R = z(x-y)$.

Choose a set of multipliers 1, 1 and 1. Then each ratio of (2) are equal to

$$\begin{aligned}
 \frac{dx + dy + dz}{x(y-z) + y(z-x) + z(x-y)} &= \frac{dx + dy + dz}{xy - xz + yz - xy + xz - yz} \\
 &= \frac{dx + dy + dz}{0}
 \end{aligned}$$

$$d(x + y + z) = 0$$

$$\begin{aligned}
 \text{Integrating, } \int d(x + y + z) &= c_1 \\
 x + y + z &= c_1. \quad \dots(3)
 \end{aligned}$$

Similarly choose another set of multipliers $\frac{1}{x}$, $\frac{1}{y}$ and $\frac{1}{z}$. Then each ratio of (2) are equal to

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{(y-z) + (z-x) + (x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

Integrating we get,

$$\int \frac{dx}{x} + \int \frac{dy}{y} + \int \frac{dz}{z} = \log c_2$$

$$\log x + \log y + \log z = \log c_2$$

$$\log xyz = \log c_2$$

$$\therefore xyz = c_2. \quad \dots(4)$$

From (3) and (4), the general solution is $\phi(x + y + z, xyz) = 0$.

Example 1.3.74. Solve : $x(y^2 + z^2)p + y(z^2 + x^2)q = z(y^2 - x^2)$.

(Apr.'14)

Solution . Given equation is $x(y^2 + z^2)p + y(z^2 + x^2)q = z(y^2 - x^2)$.

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x(y^2 + z^2)} = \frac{dy}{y(z^2 + x^2)} = \frac{dz}{z(y^2 - x^2)} \quad \dots(1)$$

Where $P = x(y^2 + z^2)$, $Q = y(z^2 + x^2)$ and $R = z(y^2 - x^2)$.

Choose a set of multipliers x , $-y$ and z . Then each ratio of (1) are equal to

$$\frac{xdx - ydy + zdz}{x^2(y^2 + z^2) - y^2(z^2 + x^2) + z^2(y^2 - x^2)} = \frac{xdx - ydy + zdz}{\begin{bmatrix} x^2y^2 + x^2z^2 - y^2z^2 \\ -x^2y^2 + z^2y^2 - z^2x^2 \end{bmatrix}}$$

$$= \frac{xdx - ydy + zdz}{0}$$

$$\therefore xdx - ydy + zdz = 0$$

$$\text{Integrating, } \int xdx - \int ydy + \int zdz = c$$

$$\frac{x^2}{2} - \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 - y^2 + z^2 = c_1 \quad [\because c_1 = 2c] \quad \dots(2)$$

Similarly choose another set of multipliers $-\frac{1}{x}$, $\frac{1}{y}$ and $\frac{1}{z}$. Then each ratio of (1) are equal to

$$\frac{-\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{-(y^2 + z^2) + (z^2 + x^2) + (y^2 - x^2)} = \frac{-\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}$$

$$\therefore -\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

Integrating we get,

$$-\int \frac{dx}{x} + \int \frac{dy}{y} + \int \frac{dz}{z} = \log c_2$$

$$-\log x + \log y + \log z = \log c_2$$

$$\log \frac{yz}{x} = \log c_2$$

$$\therefore \frac{yz}{x} = c_2. \quad \dots(3)$$

From (2) and (3), the general solution is $\phi(x^2 - y^2 + z^2, \frac{yz}{x}) = 0$.

Example 1.3.75. Solve : $x(y^2 + z)p + y(x^2 + z)q = z(x^2 - y^2)$.
(Apr.'11)

Solution . Given equation is $x(y^2 + z)p + y(x^2 + z)q = z(x^2 - y^2)$.
... (1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)} \quad \dots(2)$$

Where $P = x(y^2 + z)$, $Q = y(x^2 + z)$ and $R = z(x^2 - y^2)$.

Choose a set of multipliers x , $-y$ and -1 . Then each ratio of (2) are equal to

$$\frac{xdx - ydy - dz}{x^2(y^2 + z) - y^2(x^2 + z) - z(x^2 - y^2)} = \frac{xdx - ydy - dz}{\begin{bmatrix} x^2y^2 + x^2z - y^2x^2 \\ -y^2z - zx^2 + zy^2 \end{bmatrix}}$$

$$= \frac{xdx - ydy - dz}{0}$$

$$\therefore xdx - ydy - dz = 0$$

Integrating, $\int x dx - \int y dy - \int dz = c$

$$\frac{x^2}{2} - \frac{y^2}{2} - z = c$$

$$x^2 - y^2 - 2z = c_1 \quad [\because c_1 = 2c] \dots (3)$$

Similarly choose another set of multipliers $\frac{1}{x}$, $-\frac{1}{y}$ and $\frac{1}{z}$. Then each ratio of (2) are equal to

$$\begin{aligned} \frac{\frac{1}{x} dx - \frac{1}{y} dy + \frac{1}{z} dz}{(y^2 + z) - (x^2 + z) + (x^2 - y^2)} &= \frac{\frac{1}{x} dx - \frac{1}{y} dy + \frac{1}{z} dz}{y^2 + z - x^2 - z + x^2 - y^2} \\ &= \frac{\frac{1}{x} dx - \frac{1}{y} dy + \frac{1}{z} dz}{0} \end{aligned}$$

$$\therefore \frac{dx}{x} - \frac{dy}{y} + \frac{dz}{z} = 0$$

Integrating we get,

$$\int \frac{dx}{x} - \int \frac{dy}{y} + \int \frac{dz}{z} = \log c_2$$

$$\log x - \log y + \log z = \log c_2$$

$$\log \frac{xz}{y} = \log c_2$$

$$\therefore \frac{xz}{y} = c_2 \dots (4)$$

From (3) and (4), the general solution is $\phi(x^2 - y^2 - 2z, \frac{xz}{y}) = 0$.

Example 1.3.76. Solve : $(y - z)p - (2x + y)q = 2x + z$. (Nov.'08)

Solution . Given equation is $(y - z)p - (2x + y)q = 2x + z$. $\dots (1)$

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\begin{aligned} \frac{dx}{P} &= \frac{dy}{Q} = \frac{dz}{R} \\ \frac{dx}{y - z} &= \frac{dy}{-(2x + y)} = \frac{dz}{(2x + z)} \end{aligned} \dots (2)$$

Where $P = (y - z)$, $Q = -(2x + y)$ and $R = (2x + z)$.

Choose a set of multipliers 1, 1 and 1. Then each ratio of (2) are

$$\frac{1}{2} \log(x+y+z) = -2 \log(x-y) + 2 \log c_3$$

$$\log(x+y+z) = -2 \log(x-y) + 2 \log c_3$$

$$\log(x+y+z) + 2 \log(x-y) = 2 \log c_3$$

$$\log(x+y+z) + \log(x-y)^2 = \log c_3^2$$

$$\log(x+y+z)(x-y)^2 = \log c_4$$

$$[c_4 = c_3^2]$$

$$\therefore (x+y+z)(x-y)^2 = c_4.$$

...(8)

From the equations (4) and (8) the general solution
 $\phi\left(\frac{x-y}{y-z}, (x+y+z)(x-y)^2\right) = 0.$

Example 1.3.80 Solve : $(x^2 - yz)p + (y^2 - zx)q = (z^2 - xy).$

(Nov.'05, Apr.'08, Apr.'12)

Solution . Given equation $(x^2 - yz)p + (y^2 - zx)q = (z^2 - xy)$..(1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{(x^2 - yz)} = \frac{dy}{(y^2 - zx)} = \frac{dz}{(z^2 - xy)} \quad \dots(2)$$

Where $P = x^2 - yz$, $Q = y^2 - zx$ and $R = z^2 - xy$.

From each of (2), we write

$$\frac{dx - dy}{(x^2 - y^2) + z(x - y)} = \frac{dy - dz}{(y^2 - z^2) + x(y - z)} = \frac{dz - dx}{(z^2 - x^2) + y(z - x)}$$

$$\frac{d(x - y)}{(x - y)(x + y + z)} = \frac{d(y - z)}{(y - z)(x + y + z)} = \frac{d(z - x)}{(z - x)(x + y + z)}$$

$$\frac{d(x - y)}{x - y} = \frac{d(y - z)}{y - z} = \frac{d(z - x)}{z - x} \quad \dots(3)$$

Grouping first and second ratio of (3) we get

$$\frac{d(x - y)}{x - y} = \frac{d(y - z)}{y - z}$$

$$\begin{aligned}
 \text{Integrating, } \int \frac{d(x-y)}{x-y} &= \int \frac{d(y-z)}{y-z} + \log c_1 \\
 \log(x-y) &= \log(y-z) + \log c_1 \\
 \log \frac{x-y}{y-z} &= \log c_1 \\
 \therefore \frac{x-y}{y-z} &= c_1.
 \end{aligned} \tag{4}$$

Grouping second and third ratio of (3) we get

$$\begin{aligned}
 \frac{d(y-z)}{y-z} &= \frac{d(z-x)}{z-x} \\
 \text{Integrating, } \int \frac{d(y-z)}{y-z} &= \int \frac{d(z-x)}{z-x} + \log c_2 \\
 \log(y-z) &= \log(z-x) + \log c_2 \\
 \log \frac{y-z}{z-x} &= \log c_2 \\
 \therefore \frac{y-z}{z-x} &= c_2.
 \end{aligned} \tag{5}$$

From (4) and (5), the general solution is $\phi\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0$.

Aliter : For second set of solution we take another choice as follows

Choose multipliers x, y and z then

$$\text{Each of (2)} = \frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz} \tag{6}$$

Choose another set of multipliers 1,1,1 then

$$\text{Each of (2)} = \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \tag{7}$$

Grouping (6) and (7) we have

$$\begin{aligned}
 \frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz} &= \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \\
 \left\{ \frac{xdx + ydy + zdz}{(x+y+z)} \right\} &= \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \\
 \frac{xdx + ydy + zdz}{(x+y+z)} &= \frac{dx + dy + dz}{1} \\
 xdx + ydy + zdz &= (x+y+z)(dx + dy + dz)
 \end{aligned}$$

Integrating on both sides, we get

$$\begin{aligned}\int x dx + \int y dy + \int z dz &= \int (x + y + z)(dx + dy + dz) + c \\ \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} &= \frac{(x + y + z)^2}{2} + c_3 \\ x^2 + y^2 + z^2 &= (x + y + z)^2 + 2c_3\end{aligned}$$

$$x^2 + y^2 + z^2 - (x + y + z)^2 = 2c_3$$

$$x^2 + y^2 + z^2 - (x^2 + y^2 + z^2 + 2xy + 2yz + 2zx) = 2c_3$$

$$-2(xy + yz + zx) = 2c_3$$

$$\Rightarrow xy + yz + zx = -c_3 \dots (8)$$

From (4) and (8), the general solution is $\phi\left(\frac{x-y}{y-z}, xy + yz + zx\right) = 0$

Example 1.3.81. Solve : $(x^2 + y^2 + yz)p + (x^2 + y^2 - xz)q = z(x + y)$. (Nov. '15)

Solution . Given $(x^2 + y^2 + yz)p + (x^2 + y^2 - xz)q = z(x + y) \dots (1)$

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\therefore \frac{dx}{x^2 + y^2 + yz} = \frac{dy}{x^2 + y^2 - xz} = \frac{dz}{z(x + y)} \dots (2)$$

Where $P = x^2 + y^2 + yz$, $Q = x^2 + y^2 - xz$ and $R = z(x + y)$.

From the above ratio, we have

$$\frac{dx - dy}{x^2 + y^2 + yz - x^2 - y^2 + xz} = \frac{dz}{z(x + y)}$$

$$\frac{dx - dy}{z(x + y)} = \frac{dz}{z(x + y)}$$

$$dx - dy = dz$$

$$d(x - y - z) = 0$$

Integrating we get $\int d(x - y - z) = c_1$

$$x - y - z = c_1 \dots (3)$$

By the method of Lagrange's multipliers, from (2) we have

$$\begin{aligned}\frac{xdx + ydy}{x^3 + xy^2 + xyz + x^2y + y^3 - xyz} &= \frac{dz}{z(x+y)} \\ \frac{xdx + ydy}{x^3 + xy^2 + x^2y + y^3} &= \frac{dz}{z(x+y)} \\ \frac{xdx + ydy}{x(x^2 + y^2) + y(x^2 + y^2)} &= \frac{dz}{z(x+y)} \\ \frac{\frac{1}{2}dx^2 + \frac{1}{2}dy^2}{(x^2 + y^2)} &= \frac{dz}{z} \quad [\because d(x^2) = 2xdx] \\ \frac{dx^2 + dy^2}{(x^2 + y^2)} - 2\frac{dz}{z} &= 0\end{aligned}$$

Integrating we get, $\log \left(\frac{x^2 + y^2}{z^2} \right) = \log c_2$.

$$\Rightarrow \frac{x^2 + y^2}{z^2} = c_2 \quad \dots(4)$$

From (3) and (4), the general solution is $\phi \left(x - y - z, \frac{x^2 + y^2}{z^2} \right) = 0$.

Example 1.3.82. Solve :

$$(y - xz)p + (yz - x)q = (x + y)(x - y - z) \quad \dots(1)$$

Solution. Given $(y - xz)p + (yz - x)q = (x + y)(x - y - z)$

$$\text{Ans. } \phi(x^2 + y^2 + z^2, -x + 2y + z) = 0$$

This is Lagrange's type

$$xzq = \left(\frac{a-b}{c} \right) xy$$

$$\frac{dx}{x} = \frac{dy}{y} \text{ and } (ax, by, cz)$$

1.4 Linear Partial Differential Equations of Second and Higher Order With Constant Coefficients

Linear partial differential equations of higher order can be classified as follows

1. The partial derivatives are occurring in the equation are all of same order (degree is 1) and the constant coefficients are called homogeneous linear partial differential equations with constant coefficients.
2. The partial derivatives are occurring in the equation are not of same order and the constant coefficients are called non homogeneous linear partial differential equations with constant coefficients.

1.4.1 Homogeneous Linear Higher Order PDE

A linear homogeneous PDE of n^{th} order with constant coefficients is given by

$$a_0 \frac{\partial^n z}{\partial x^n} + a_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + a_2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots + a_n \frac{\partial^n z}{\partial y^n} = g(x, y) \quad \dots(1)$$

Where $a_0, a_1, a_2, \dots, a_n$ are constants.

Equation (1) can be written as

$$[a_0 D^n + a_1 D^{n-1} D' + a_2 D^{n-2} (D')^2 + \dots + a_n (D')^n] z = g(x, y) \quad \dots(2)$$

Where $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$.

Also equation (2) is shortly written as $f(D, D')z = g(x, y)$.

Thus the solution of (2) is $z = \text{C.F.} + \text{P.I.}$

To find Complementary Function (C.F.)

In (2), replace D by m , D' by 1 and equating to zero we get the auxiliary of (2) as follows

$$a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0 \quad \dots(3)$$

Solving (3) we get n roots namely $m_1, m_2, m_3, \dots, m_n$

Case 1: If the roots are real or complex and distinct, say $m_1 \neq m_2 \neq m_3 \neq \dots \neq m_n$ then the complementary function is $z = f_1(y + m_1x) + f_2(y + m_2x) + f_3(y + m_3x) + \dots + f_n(y + m_nx)$

Case 2: If any two roots are equal, say $m_1 = m_2 = m$ and others are distinct, then the complementary function is $z = f_1(y + mx) + xf_2(y + mx) + f_3(y + m_3x) + \dots + f_n(y + m_nx)$

Case 3: If any three roots are equal, say $m_1 = m_2 = m_3 = m$ and others are distinct, then the complementary function is $z = f_1(y + mx) + xf_2(y + mx) + x^2f_3(y + mx) + \dots + f_n(y + m_nx)$

Method of Finding Particular Integral (P.I.)

Let the given PDE is of the form $f(D, D')z = g(x, y)$. Then the P.I. is

$$P.I. = \frac{1}{f(D, D')} \times g(x, y)$$

Type 1: If $g(x, y) = e^{ax+by}$ then

$$P.I. = \frac{1}{f(D, D')} \times g(x, y)$$

Replace D by a and D' by b we get

$$= \frac{1}{f(a, b)} \times g(x, y) \text{ provided } f(a, b) \neq 0$$

Note : If $f(a, b) = 0$ then multiply the numerator by x and differentiate the denominator partially w.r. to D then replace D by a and D' by b , provided the $f_x(a, b) \neq 0$. Suppose that, if $f_x(a, b) = 0$ then proceed the above step repeatedly, until the differential operators in the denominator are eliminated.

Type 2: If $g(x, y) = \sin(ax + by)$ or $\cos(ax + by)$ then

$$P.I. = \frac{1}{f(D, D')} \times \sin(ax + by) \text{ or } \cos(ax + by)$$

Replace D^2 by $-a^2$, D'^2 by $-b^2$ and DD' by $-ab$ we get

$$= \frac{1}{f(-a^2, -ab, -b^2)} \times \sin(ax + by) \text{ or } \cos(ax + by)$$

provided $f(-a^2, -ab, -b^2) \neq 0$

Type 3: If $g(x, y) = x^m y^n$, m, n are positive integers then

$$\begin{aligned} P.I. &= \frac{1}{f(D, D')} \times x^m y^n \\ &= [f(D, D')]^{-1} x^m y^n \end{aligned}$$

Expand $[f(D, D')]^{-1}$ in ascending powers of D and D' by binomial expansion upto D^m and D'^n .

Note : In the polynomial expression $x^m y^n$,

1. If $m < n$ then write $f(D, D')$ as in the form $f\left(\frac{D}{D'}\right)$ then apply binomial series expansion.
2. If $m > n$ then write $f(D, D')$ as in the form $f\left(\frac{D'}{D}\right)$ then apply binomial series expansion.

Type 4: If $g(x, y) = e^{ax+by} R(x, y)$ Where $R(x, y)$ is some function of x, y .

$$\begin{aligned} P.I. &= \frac{1}{f(D, D')} \times e^{ax+by} R(x, y) \\ &= e^{ax+by} \frac{1}{f(D+a, D'+b)} \times R(x, y) \end{aligned}$$

If $R(x, y)$ is an exponential function in x and y then use Type 1 and proceed or $R(x, y)$ is a trigonometric function in x and y then use Type 2 and proceed.

Type 5: If $g(x, y) =$ anything other than in Type 1, 2, 3, 4 then the P.I. as follows

$$P.I. = \frac{1}{f(D, D')} \times g(x, y)$$

Resolve the denominator $f(D, D')$ into linear factors of the form $(D - m_1 D')(D - m_2 D')$.

$$\text{i.e., } P.I. = \frac{1}{(D - m_1 D')(D - m_2 D')} \times g(x, y)$$

Leave the factor $(D - m_1 D')$ and change y to $c - m_1 x$ in $f(x, y)$ and integrate w.r.to x .

$$\text{i.e., } P.I. = \frac{1}{(D - m_1 D')} \int g(x, c - m_1 x) dx$$

After integration c is to be replaced by $y + m_1 x$. Also repeat the same procedure for each linear factor in the denominator.

Example 1.4.1. Solve : $(D^2 - 5DD' + 6D'^2)z = 0$

Solution . Given equation is, $(D^2 - 5DD' + 6D'^2)z = 0$ (1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 5m + 6 = 0$$

$$(m - 2)(m - 3) = 0$$

$$\Rightarrow m = 2, 3$$

The required solution is,

$$z = f_1(y + m_1 x) + f_2(y + m_2 x)$$

$$\text{i.e., } z = f_1(y + 2x) + f_2(y + 3x).$$

Example 1.4.2. Solve : $(D^2 - 6DD' + 9D'^2)z = 0$

Nov.'14

Solution . Given equation is, $(D^2 - 6DD' + 9D'^2)z = 0$ (1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 6m + 9 = 0$$

$$(m - 3)^2 = 0$$

$$\Rightarrow m = 3, 3$$

The required solution is,

$$z = f_1(y + mx) + xf_2(y + mx)$$

$$\text{i.e., } z = f_1(y + 3x) + xf_2(y + 3x).$$

Example 1.4.3. Solve : $(D^2 + 4D'^2)z = 0$

Solution . Given equation is, $(D^2 + 4D'^2)z = 0$.

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 + 4 = 0$$

$$m^2 = -4 = 4i^2$$

$$\Rightarrow m = \pm i2$$

The required solution is,

$$z = f_1(y + m_1x) + f_2(y + m_2x)$$

$$\text{i.e., } z = f_1(y + i2x) + f_2(y - i2x)$$

Example 1.4.4. Solve : $(D^3 - 4D^2D' + 4DD'^2)z = 0$

Solution . Given equation is, $(D^3 - 4D^2D' + 4DD'^2)z = 0$ (1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^3 - 4m^2 + 4m = 0$$

$$m(m^2 - 4m + 4) = 0$$

$$m(m - 2)^2 = 0$$

$$\Rightarrow m = 0, 2, 2$$

The required solution is,

$$z = f_1(y + m_1x) + f_2(y + mx) + xf_3(y + mx)$$

$$\text{i.e., } z = f_1(y + 0x) + f_2(y + 2x) + xf_3(y + 2x).$$

Example 1.4.5. Solve : $(D^4 - D'^4)z = 0$

$$4. (D^4 - 16D'^4)z = 0.$$

$$\text{Ans. } z = f_1(y + 2x) + f_2(y - 2x) + f_3(y + 2ix) + f_4(y - 2ix)$$

$$5. (D^3 - D^2D' + DD'^2 - D'^3)z = 0.$$

$$\text{Ans. } z = f_1(y + x) + f_2(y + ix) + xf_3(y - ix)$$

Type 1: If $g(x, y) = e^{ax+by}$ then replace D by a and D' by b in P.I.

Example 1.4.7. Find P.I. of $(D^2 + 2DD' + D'^2)z = e^{2x+y}$

Solution .

$$P.I. = \frac{1}{D^2 + 2DD' + D'^2} e^{2x+y}$$

Replace D by 2 and D' by 1 we get

$$= \frac{1}{2^2 + 2 \cdot 2 \cdot 1 + 1^2} e^{2x+y}$$

$$\therefore P.I. = \frac{1}{9} e^{2x+y}$$

Example 1.4.8. Find P.I. of $(D^2 + 2DD')z = e^x$

Solution .

$$P.I. = \frac{1}{D^2 + 2DD'} e^{x+0y}$$

Replace D by 1 and D' by 0 we get

$$= \frac{1}{1^2 + 2 \cdot 1 \cdot 0} e^{x+0y} = \frac{1}{1} e^x$$

$$\therefore P.I. = e^x.$$

Example 1.4.9. Find P.I. of $(D^2 - 3DD' + 2D'^2)z = e^{2x+y}$

Solution .

$$P.I. = \frac{1}{D^2 - 3DD' + 2D'^2} e^{2x+y}$$

Replace D by 2 and D' by 1 we get

$$= \frac{1}{4 - 6 + 2} e^{2x+y} = \frac{1}{0} e^{2x+y}$$

$$= \frac{x}{2D - 3D'} e^{2x+y} \quad \text{Replace } D \text{ by 2 and } D' \text{ by 1}$$

$$= \frac{x}{4 - 3} e^{2x+y}$$

$$\therefore P.I. = x e^{2x+y}.$$

Example 1.4.10. Solve : $(D^3 - 3D^2D' + 4D'^4)z = e^{x+2y}$

Solution . Given equation is, $(D^3 - 3D^2D' + 4D'^3)z = e^{x+2y}$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$\begin{aligned} m^3 - 3m^2 + 4 &= 0 \\ (m+1)(m-2)^2 &= 0 \\ \Rightarrow m &= -1, 2, 2 \end{aligned}$$

The complementary function (C.F.) is

$$\begin{aligned} z &= f_1(y + m_1x) + f_2(y + mx) + xf_3(y + mx) \\ \text{i.e., } z &= f_1(y - x) + f_2(y + 2x) + xf_3(y + 2x) \end{aligned}$$

$$P.I. = \frac{1}{D^3 - 3D^2D' + 4D'^3} e^{x+2y}$$

Replace D by 1 and D' by 2 we get

$$= \frac{1}{1 - 6 + 32} e^{2x+y}$$

$$\therefore P.I. = \frac{1}{27} e^{x+2y}$$

Thus the complete solution is,

$$\begin{aligned} z &= \text{C.F.} + \text{P.I.} \\ \therefore z &= f_1(y - x) + f_2(y + 2x) + xf_3(y + 2x) + \frac{1}{27} e^{x+2y} \end{aligned}$$

Example 1.4.11. Solve : $(D^2 - 4DD' + 4D'^2)z = e^{2x+y}$

Solution . Given equation is, $(D^2 - 4DD' + 4D'^2)z = e^{2x+y}$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$\begin{aligned} m^2 - 4m + 4 &= 0 \\ (m-2)^2 &= 0 \\ \Rightarrow m &= 2, 2 \end{aligned}$$

$$\begin{array}{c} \nearrow \quad \searrow \\ -2 \quad -2 \\ \hline (m-2)^2 = 0 \end{array}$$

$$= \frac{2}{81} \text{I.P. } i [\cos(3x + 6y) + i \sin(3x + 6y)] \quad [\because i^2 = -1]$$

$$\therefore P.I. = \frac{2}{81} \cos(3x + 6y).$$

Method 2:

$$P.I. = \frac{1}{D^3 - 4D^2D' + 4DD'^2} 6 \sin(3x + 6y)$$

Replace $D^2 = -9$, $D'^2 = -36$ and $DD' = -18$

$$= 6 \frac{1}{-9D - 4(-9)D' + 4(-36)D} \sin(3x + 6y)$$

$$= 6 \frac{1}{-9D - 4(-9)D' - 144D} \sin(3x + 6y)$$

$$= 6 \frac{1}{-153D + 36D'} \sin(3x + 6y)$$

$$= \frac{6}{9} \frac{1}{4D' - 17D} \sin(3x + 6y) \quad \frac{2}{9}$$

$$= \frac{2}{3} \frac{D}{(4DD' - 17D^2)} \sin(3x + 6y)$$

$$= \frac{2}{3} \frac{D}{4(-18) - 17(-9)} \sin(3x + 6y)$$

$$= \frac{2}{3} \frac{D \sin(3x + 6y)}{-72 + 153} = \frac{2}{3} \frac{3 \cos(3x + 6y)}{81}$$

$$\therefore P.I. = \frac{2}{81} \cos(3x + 6y).$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1y + f_2(y + 2x) + xf_3(y + 2x) + \frac{2}{81} \cos(3x + 6y).$$

Example 1.4.13. Solve : $(2D^2 - 5DD' + 2D'^2)z = 5 \sin(2x + y)$
(Nov. '13)

Solution . Given $(2D^2 - 5DD' + 2D'^2)z = 5 \sin(2x + y)$.

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$2m^2 - 5m + 2 = 0$$

$$(2m - 1)(m - 2) = 0. \Rightarrow m = 2, \frac{1}{2}$$

The complementary function (C.F.) is

$$z = f_1(y + m_1x) + f_2(y + m_2x)$$

$$\text{i.e., } z = f_1(y + 2x) + f_2\left(y + \frac{1}{2}x\right)$$

To find P.I. : Method 1

$$P.I. = \frac{1}{2D^2 - 5DD' + 2D'^2} 5 \sin(2x + y)$$

$$= 5 \text{ I.P. } \frac{1}{2D^2 - 5DD' + 2D'^2} e^{i(2x+y)}$$

Replace D by $2i$ and D' by i we get

$$= 5 \text{ I.P. } \frac{1}{2(2i)^2 - 5(2i)(i) + 2(i)^2} e^{i(2x+y)}$$

$$= 5 \text{ I.P. } \frac{1}{2(-4) - 5(-2) + 2(-1)} e^{i(2x+y)}$$

$$= 5 \text{ I.P. } \frac{1}{-8 + 10 - 2} e^{i(2x+y)} = 5 \text{ I.P. } \frac{1}{0} e^{i(2x+y)}$$

$$= 5 \text{ I.P. } \frac{x}{4D - 5D'} e^{i(2x+y)}$$

$$= 5 \text{ I.P. } \frac{x}{4(2i) - 5(i)} e^{i(2x+y)}$$

$$= 5 \text{ I.P. } \frac{x}{3i} e^{i(2x+y)}$$

$$= \frac{5}{3} \text{ I.P. } (-ix) [\cos(2x + y) + i \sin(2x + y)]$$

$$\therefore P.I. = -\frac{5}{3} x \cos(2x + y)$$

Method 2:

$$P.I. = \frac{1}{2D^2 - 5DD' + 2D'^2} 5 \sin(2x + y)$$

Replace $D^2 = -4$, $D'^2 = -1$ and $DD' = -2$

$$= 5 \frac{1}{2(-4) - 5(-2) + 2(-1)} \sin(2x + y)$$

$$\begin{aligned}
 &= \frac{5}{0} \sin(2x + y) \\
 &= 5 \frac{x}{4D - 5D'} \sin(2x + y) \\
 &= 5x \frac{D}{4D^2 - 5DD'} \sin(2x + y) \\
 &= 5x \frac{D \sin(2x + y)}{4(-4) - 5(-2)} \\
 &= 5x \frac{2 \cos(2x + y)}{-16 + 10} \\
 &= -\frac{10x}{6} \cos(2x + y)
 \end{aligned}$$

$$\therefore P.I. = -\frac{5}{3} x \cos(2x + y).$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y + 2x) + f_2\left(y + \frac{1}{2}x\right) - \frac{5}{3} \cos(2x + y).$$

Example 1.4.14. Solve : $(D^3 + D^2D' + 4DD'^2 + 4D'^3)z = \cos(2x + y)$
(Apr. '12)

Solution . Given equation is,

$$(D^3 + D^2D' + 4DD'^2 + 4D'^3)z = \cos(2x + y). \quad \dots(1)$$

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^3 + m^2 + 4m + 4 = 0$$

$$(m + 1)(m^2 + 4) = 0$$

$$\Rightarrow m = -1, \pm 2i \quad (\text{or}) \quad m = -1, 2i, -2i$$

The complementary function (C.F.) is

$$\begin{aligned}
 z &= f_1(y + m_1x) + f_2(y + m_2x) + f_3(y + m_3x) \\
 \text{i.e., } z &= f_1(y - x) + f_2(y + 2ix) + f_3(y - 2ix)
 \end{aligned}$$

To find P.I. : Method 1

$$\begin{aligned}
 P.I. &= \frac{1}{D^3 + D^2 D' + 4DD'^2 + 4D'^3} \cos(2x + y) \\
 &= \text{R.P.} \frac{1}{D^3 + D^2 D' + 4DD'^2 + 4D'^3} e^{i(2x+y)} \\
 &\quad \text{Replace } D \text{ by } 2i \text{ and } D' \text{ by } i \text{ we get} \\
 &= \text{R.P.} \frac{1}{(2i)^3 + (2i)^2(i) + 4(2i)(i)^2 + 4(i)^3} e^{i(2x+y)} \\
 &= \text{R.P.} \frac{1}{i^3[8 + 4 + 8 + 4]} e^{i(2x+y)} \\
 &= \text{R.P.} \frac{i}{24} e^{i(2x+y)} \\
 &= \frac{1}{24} \text{R.P. } i [\cos(2x + y) + i \sin(2x + y)] \\
 \therefore P.I. &= -\frac{1}{24} \sin(2x + y)
 \end{aligned}$$

Method 2:

$$\begin{aligned}
 P.I. &= \frac{1}{D^3 + D^2 D' + 4DD'^2 + 4D'^3} \cos(2x + y) \\
 &\quad \text{Replace } D^2 = -4, D'^2 = -1 \text{ and } DD' = -2 \\
 &= \frac{1}{-4D - 4D' - 4D - 4D'} \cos(2x + y) \\
 &= -\frac{1}{8} \frac{1}{(D + D')} \cos(2x + y) \\
 &= -\frac{1}{8} \frac{D}{(D^2 + DD')} \cos(2x + y) \\
 &= -\frac{1}{8} \frac{D \cos(2x + y)}{(-4 - 2)} \\
 &= \frac{1}{8} \frac{-2 \sin(2x + y)}{6} \\
 \therefore P.I. &= -\frac{1}{24} \sin(2x + y)
 \end{aligned}$$

Thus the complete solution is,

$$\begin{aligned}
 z &= \text{C.F.} + \text{P.I.} \\
 \therefore z &= f_1(y - x) + f_2(y + 2ix) + f_3(y - 2ix) - \frac{1}{24} \sin(2x + y).
 \end{aligned}$$

Example 1.4.16. Solve : $(D^3 - 7DD'^2 - 6D'^3)z = e^{2x+y} + \sin(x+2y)$

(Nov.'09)

Solution . Given $(D^3 - 7DD'^2 - 6D'^3)z = e^{2x+y} + \sin(x+2y) \dots (1)$

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^3 - 7m - 6 = 0$$

$$(m+1)(m^2 - m - 6) = 0$$

$$(m+1)(m+2)(m-3) = 0$$

$$\Rightarrow m = -1, -2, 3$$

The complementary function (C.F.) is

$$z = f_1(y + m_1x) + f_2(y + m_2x) + f_3(y + m_3x)$$

$$\text{i.e., } z = f_1(y - x) + f_2(y - 2x) + f_3(y + 3x)$$

$$P.I._1 = \frac{1}{D^3 - 7DD'^2 - 6D'^3} e^{2x+y}$$

Replace D by 2 and D' by 1 we get

$$= \frac{1}{8 - 14 - 6} e^{2x+y}$$

$$\therefore P.I._1 = -\frac{1}{12} e^{2x+y}.$$

$$P.I._2 = \frac{1}{D^3 - 7DD'^2 - 6D'^3} \sin(x+2y)$$

Replace $D^2 = -1$, $D'^2 = -4$ and $DD' = -2$

$$= \frac{1}{-D + 28D - 6(-4)D'} \sin(x+2y)$$

$$= \frac{1}{27D + 24D'} \sin(x+2y) = \frac{1}{3} \frac{1}{9D + 8D'} \sin(x+2y)$$

$$= \frac{1}{3} \frac{D}{9D^2 + 8DD'} \sin(x+2y)$$

$$= \frac{1}{3} \frac{D \sin(x+2y)}{-9 - 16}$$

$$\therefore P.I._2 = -\frac{1}{75} \cos(x+2y).$$

Type 3: If $g(x, y) = x^m y^n$, m, n are positive integers.

Note :

1. $(1 - x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$
2. $(1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$
3. $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots$
4. $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - \dots$

Example 1.4.17. Solve : $(D^2 - 2DD')z = e^{2x} + x^3y$ (Nov.'14)

Solution . Given $(D^2 - 2DD')z = x^3y$(1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$\begin{aligned} m^2 - 2m &= 0 \\ m(m - 1) &= 0 \\ \Rightarrow m &= 0, 1. \end{aligned}$$

The complementary function (C.F.) is

$$\begin{aligned} z &= f_1(y + m_1x) + f_2(y + m_2x) \\ \text{i.e., } z &= f_1(y + 0x) + f_2(y + x) \end{aligned}$$

$$P.I._1 = \frac{1}{D^2 - 2DD'} e^{2x+0y}$$

Replace D by 2 and D' by 0 we get

$$= \frac{1}{4 - 0} e^{2x+0y}$$

$$\therefore P.I._1 = \frac{1}{4} e^{2x}.$$

$$P.I._2 = \frac{1}{D^2 - 2DD'} x^3y = \frac{1}{D^2 \left[1 - \frac{2D'}{D}\right]} x^3y$$

$$= \frac{1}{D^2} \left[1 - \frac{2D'}{D}\right]^{-1} x^3y$$

$$\begin{aligned}
&= \frac{1}{D^2} \left[1 + \frac{2D'}{D} + \frac{4D'^2}{D^2} + \dots \right] x^3 y \\
&= \frac{1}{D^2} \left[x^3 y + \frac{2D'}{D} x^3 y + \frac{4D'^2}{D^2} x^3 y + \dots \right] \\
&= \frac{1}{D^2} \left[x^3 y + \frac{2}{D} x^3 + 0 \right] \\
&= \frac{1}{D^2} x^3 y + \frac{2}{D^3} x^3 \\
&= \frac{x^5 y}{4 \times 5} + \frac{2x^6}{4 \times 5 \times 6} \\
\therefore P.I._2 &= \frac{x^5 y}{20} + \frac{x^6}{60}
\end{aligned}$$

D = 1

Thus the complete solution is,

$$\begin{aligned}
z &= C.F. + P.I._1 + P.I._2 \\
\therefore z &= f_1(y + 0x) + f_2(y + x) + \frac{1}{4} e^{2x} + \frac{x^5 y}{20} + \frac{x^6}{60}.
\end{aligned}$$

Example 1.4.18. Solve : $(D^2 + 2DD' + D'^2)z = x^2 + xy - y^2$

Solution . Given $(D^2 + 2DD' + D'^2)z = x^2 + xy - y^2$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$\begin{aligned}
m^2 + 2m + 1 &= 0 \\
(m + 1)^2 &= 0 \\
\Rightarrow m &= -1, -1.
\end{aligned}$$

The complementary function (C.F.) is

$$\begin{aligned}
z &= f_1(y + mx) + x f_2(y + mx) \\
\text{i.e., } z &= f_1(y - x) + x f_2(y - x)
\end{aligned}$$

$$\begin{aligned}
P.I. &= \frac{1}{D^2 + 2DD' + D'^2} (x^2 + xy - y^2) \\
&= \frac{1}{D^2 \left[1 + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) \right]} (x^2 + xy - y^2)
\end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{D^2} \left[1 + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) \right]^{-1} (x^2 + xy - y^2) \\
 &= \frac{1}{D^2} \left[1 - \frac{2D'}{D} - \frac{D'^2}{D^2} + \frac{4D'^2}{D^2} + \dots \right] (x^2 + xy - y^2) \\
 &= \frac{1}{D^2} \left[1 - \frac{2D'}{D} + \frac{3D'^2}{D^2} + \dots \right] (x^2 + xy - y^2) \\
 &= \frac{1}{D^2} \left[(x^2 + xy - y^2) - \frac{2}{D}(x - 2y) - \frac{3}{D^2}(-2) + 0\dots \right] \\
 &= \frac{1}{D^2} \left[(x^2 + xy - y^2) - 2 \left(\frac{x^2}{2} - 2xy \right) - 3 \left(-\frac{2x^2}{2} \right) \right] \\
 &= \frac{1}{D^2} [5xy - y^2 + 3x^2] \\
 &= \frac{5x^3y}{6} - \frac{x^2y^2}{2} + \frac{3x^4}{12} \\
 \therefore P.I. &= \frac{5x^3y}{6} - \frac{x^2y^2}{2} + \frac{x^4}{4}.
 \end{aligned}$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y - x) + xf_2(y - x) + \frac{5x^3y}{6} - \frac{x^2y^2}{2} + \frac{x^4}{4}.$$

Example 1.4.19. Solve : $(D^2 + 2DD' + D'^2)z = e^{x-y} + x^2y$ (Nov.'11)

Solution . Given $(D^2 + 2DD' + D'^2)z = e^{x-y} + x^2y$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 + 2m + 1 = 0$$

$$(m + 1)^2 = 0. \Rightarrow m = -1, -1.$$

The complementary function (C.F.) is

$$z = f_1(y + mx) + xf_2(y + mx)$$

$$\text{i.e., } z = f_1(y - x) + xf_2(y - x)$$

$$P.I._1 = \frac{1}{D^2 + 2DD' + D'^2} e^{x-y}$$

Replace D by 1 and D' by -1 we get

$$= \frac{1}{1 - 2 + 1} e^{x-y} = \frac{1}{0} e^{x-y}$$

$$= \frac{x}{2D + 2} e^{x-y}$$

Replace D by 1 and D' by -1 we get

$$\therefore P.I._1 = \frac{x}{4} e^{x-y}$$

$$P.I._2 = \frac{1}{D^2 + 2DD' + D'^2} x^2 y$$

$$= \frac{1}{D^2 \left[1 + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) \right]} x^2 y$$

$$= \frac{1}{D^2} \left[1 + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) \right]^{-1} x^2 y$$

$$= \frac{1}{D^2} \left[1 - \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right)^2 \dots \right] x^2 y$$

$$= \frac{1}{D^2} \left[x^2 y - \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) x^2 y + 0 \right]$$

$$= \frac{1}{D^2} \left[x^2 y - \frac{2D'}{D} x^2 y - \frac{D'^2}{D^2} x^2 y + 0 \right]$$

$$= \frac{1}{D^2} \left[x^2 y - \frac{2}{D} x^2 - 0 \right]$$

$$= \frac{1}{D^2} x^2 y - \frac{2}{D^3} x^2$$

$$= \frac{x^4 y}{3 \times 4} - \frac{2x^5}{3 \times 4 \times 5}$$

$$\therefore P.I._2 = \frac{x^4 y}{12} - \frac{2x^5}{60}$$

Thus the complete solution is,

$$z = \text{C.F.} + P.I._1 + P.I._2$$

$$\therefore z = f_1(y-x) + x f_2(y-x) + \frac{x}{4} e^{x-y} + \frac{x^4 y}{12} - \frac{2x^5}{60}$$

Example 1.4.20. Solve : $(D^2 - D')z = 0$

Solution. Given $(D^2 - D')z = 0$

In (1), replace D by m and D' by m'

auxiliary equation is $m^2 - 1 = 0$

$$m^2 - 1 = 0$$

$$\Rightarrow m = \pm 1$$

The complementary function (C.F.) is

$$z = f_1(y-x) + f_2(y+x)$$

$$\text{i.e., } z = f_1(y-x) + f_2(y+x)$$

$$P.I._1 = \frac{1}{D^2 - D'} e^{x-y}$$

Replace D by 1 and D' by -1

$$= \frac{1}{1 - 1} e^{x-y} = \frac{1}{0} e^{x-y}$$

$$= \frac{x}{2D} e^{x-y} \text{ Replace } D \text{ by } 1$$

$$\therefore P.I._1 = \frac{x}{2} e^{x-y}$$

$$P.I._2 = \frac{1}{D^2 - D'} x^2 y$$

$$= \frac{1}{D^2} \left[1 + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) \right]^{-1} x^2 y$$

$$= \frac{1}{D^2} \left[1 - \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right)^2 \dots \right] x^2 y$$

$$= \frac{1}{D^2} \left[x^2 y - \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right) x^2 y + 0 \right]$$

$$= \frac{1}{D^2} \left[x^2 y - \frac{2D'}{D} x^2 y - \frac{D'^2}{D^2} x^2 y + 0 \right]$$

$$= \frac{1}{D^2} \left[x^2 y - \frac{2}{D} x^2 - 0 \right]$$

$$= \frac{1}{D^2} x^2 y - \frac{2}{D^3} x^2$$

$$\therefore P.I._2 = \frac{x^4 y}{12} - \frac{2x^5}{60}$$

Example 1.4.20. Solve : $(D^2 - D'^2)z = e^{x-y} + x^2y$ (Nov.'10)

Solution . Given $(D^2 - D'^2)z = e^{x-y} + x^2y$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 - 1 = 0$$

$$m^2 = 1$$

$$\Rightarrow m = 1, -1.$$

The complementary function (C.F.) is

$$z = f_1(y + m_1x) + f_2(y + m_2x)$$

$$\text{i.e., } z = f_1(y + x) + f_2(y - x)$$

$$P.I._1 = \frac{1}{D^2 - D'^2} e^{x-y}$$

Replace D by 1 and D' by -1 we get

$$= \frac{1}{1-1} e^{x-y} = \frac{1}{0} e^{x-y}$$

$$= \frac{x}{2D} e^{x-y} \quad \text{Replace } D \text{ by 1 and } D' \text{ by } -1 \text{ we get}$$

$$\therefore P.I._1 = \frac{x}{2} e^{x-y}.$$

$$P.I._2 = \frac{1}{D^2 - D'^2} x^2y$$

$$= \frac{1}{D^2 \left[1 - \frac{D'^2}{D^2} \right]} x^2y$$

$$= \frac{1}{D^2} \left[1 - \frac{D'^2}{D^2} \right]^{-1} x^2y$$

$$= \frac{1}{D^2} \left[1 + \frac{D'^2}{D^2} + \frac{D'^4}{D^4} + \dots \right] x^2y$$

$$= \frac{1}{D^2} \left[x^2y + \frac{D'^2}{D^2} x^2y + \frac{D'^4}{D^4} x^2y + \dots \right]$$

$$= \frac{1}{D^2} [x^2y + 0]$$

$$\therefore P.I._2 = \frac{x^4y}{12}.$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}_1 + \text{P.I.}_2$$

$$\therefore z = f_1(y+x) + f_2(y-x) + \frac{x}{2} e^{x-y} + \frac{x^4 y}{12}.$$

Example 1.4.21. Solve : $(D^2 + 4DD' - 5D'^2)z = xy + \sin(2x + 3y)$ (Apr. '13)

Solution . Given $(D^2 + 4DD' - 5D'^2)z = xy + \sin(2x + 3y)$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 + 4m - 5 = 0$$

$$(m-1)(m+5) = 0$$

$$\Rightarrow m = 1, -5.$$

The complementary function (C.F.) is

$$z = f_1(y + m_1 x) + f_2(y + m_2 x)$$

i.e., $z = f_1(y + x) + f_2(y - 5x)$

$$\begin{aligned} P.I._1 &= \frac{1}{D^2 + 4DD' - 5D'^2} xy \\ &= \frac{1}{D^2 \left[1 + \left(\frac{4D'}{D} - \frac{5D'^2}{D^2} \right) \right]} xy \\ &= \frac{1}{D^2} \left[1 + \left(\frac{4D'}{D} - \frac{5D'^2}{D^2} \right) \right]^{-1} xy \\ &= \frac{1}{D^2} \left[1 - \left(\frac{4D'}{D} + \frac{D'^2}{D^2} \right) + \left(\frac{2D'}{D} + \frac{D'^2}{D^2} \right)^2 \dots \right] xy \\ &= \frac{1}{D^2} \left[xy - \frac{4D'}{D} xy - \frac{D'^2}{D^2} xy + 0 \right] \\ &= \frac{1}{D^2} \left[xy - \frac{4}{D} x - 0 \right] \\ &= \frac{1}{D^2} xy - \frac{4}{D^3} x \end{aligned}$$

$$= \frac{x^3 y}{2 \times 3} - \frac{4x^4}{2 \times 3 \times 4} = \frac{x^3 y}{6} - \frac{x^4}{6}$$

$$\therefore P.I._1 = \frac{x^3}{6}(y-x).$$

$$P.I._2 = \frac{1}{D^2 + 4DD' - 5D'^2} \sin(2x + 3y)$$

Replace $D^2 = -4$, $D'^2 = -9$ and $DD' = -6$

$$= \frac{1}{-4 + 4(-6) - 5(-9)} \sin(2x + 3y)$$

$$= \frac{1}{-4 - 24 + 45} \sin(2x + 3y)$$

$$\therefore P.I._2 = \frac{1}{17} \sin(2x + 3y)$$

Thus the complete solution is,

$$z = \text{C.F.} + P.I._1 + P.I._2$$

$$\therefore z = f_1(y+x) + f_2(y-5x) + \frac{x^3}{6}(y-x) + \frac{1}{17} \sin(2x+3y).$$

Example 14.22. Solve : $(D^3 - 2D^2D')z = \sin(x+2y) + xy$ (Nov.'11)

Solution . Given $(D^3 - 2D^2D')z = \sin(x+2y) + xy$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^3 - 2m^2 = 0$$

$$m^2(m-2) = 0$$

$$\Rightarrow m = 0, 0, 2.$$

The complementary function (C.F.) is

$$z = f_1(y+mx) + xf_2(y+mx) + f_3(y+m_3x)$$

$$\text{i.e., } z = f_1(y+0x) + xf_2(y+0x) + f_3(y-2x)$$

$$P.I._1 = \frac{1}{D^3 - 2D^2D'} \sin(x+2y) \text{ Replace } D^2 = -1, DD' = -2$$

$$= \frac{1}{-D + 2D'} \sin(x+2y)$$

$$= \frac{D}{-D^2 + 2DD'} \sin(x + 2y) = \frac{1}{1 - 4} \sin(x + 2y)$$

$$\therefore P.I._1 = -\frac{1}{3} \sin(x + 2y).$$

$$P.I._2 = \frac{1}{D^3 - 2D^2D'} xy = \frac{1}{D^3 \left[1 - \frac{2D'}{D}\right]} xy$$

$$= \frac{1}{D^3} \left[1 - \frac{2D'}{D}\right]^{-1} xy$$

$$= \frac{1}{D^3} \left[1 + \frac{2D'}{D} + \frac{4D'^2}{D^2} + \dots\right] xy$$

$$= \frac{1}{D^3} \left[xy + \frac{2D'}{D} xy + \frac{4D'^2}{D^2} xy + \dots\right]$$

$$= \frac{1}{D^3} \left[xy + \frac{2}{D} x + 0\right] = \frac{1}{D^3} xy + \frac{2}{D^4} x$$

$$= \frac{x^4 y}{2 \times 3 \times 4} + \frac{2x^5}{2 \times 3 \times 4 \times 5}$$

$$\therefore P.I._2 = \frac{x^4 y}{24} + \frac{x^5}{60}$$

Thus the complete solution is,

$$z = \text{C.F.} + P.I._1 + P.I._2$$

$$\therefore z = f_1(y) + x f_2(y) + f_3(y - 2x) - \frac{1}{3} \sin(x + 2y) + \frac{x^4 y}{24} + \frac{x^5}{60}$$

Exercise

Solve the following equations :

1. $(D^2 + 2DD' + D'^2)z = x^2 y + e^{x-y}.$

Ans. $z = f_1(y - x) + x f_2(y - x) + \frac{x^4 y}{12} - \frac{x^5}{30} + \frac{x^2}{2} e^{x-y}$

2. $(D^2 - 6DD' + 5D'^2)z = \sinh(x + y) + xy.$

Ans. $z = f_1(y + x) + f_1(y + 5x) - \frac{x}{8} e^{x+y} - \frac{x}{8} e^{-(x+y)} + \frac{x^3 y}{6} + \frac{x^4}{4}$

3. $(D^2 - 6DD' + 5D'^2)z = e^x \sinh y + xy.$

Ans. $z = f_1(y + x) + f_2(y + 5x) - \frac{1}{8} x e^{x+y} - \frac{1}{24} e^{x-y} + \frac{x^3 y}{6} + \frac{x^4}{4}$

4. $(D^2 + DD' - 6D'^2)z = x^2 y + e^{3x+y}.$

Ans. $z = f_1(y + 2x) + f_2(y - 3x) + \frac{x^4 y}{24} - \frac{x^5}{60} + \frac{e^{3x+y}}{6}$

Type 4: If $g(x, y) = e^{ax+by} R(x, y)$ Where $R(x, y)$ is exponential or trigonometric function of x and y .

Example 1.4.13. Solve : $(D^3 + D^2 D' - DD'^2 - D'^3)z = e^x \cos 2y$
(Nov.'09, Apr.'10)

Solution . Given $(D^3 + D^2 D' - DD'^2 - D'^3)z = e^x \cos 2y$ (1)

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^3 + m^2 - m - 1 = 0$$

$$m^2(m+1) - 1(m+1) = 0$$

$$(m+1)(m^2-1) = 0$$

$$\therefore m = -1, \pm 1 = 1, -1, -1.$$

The complementary function (C.F.) is

$$z = f_1(y + m_1 x) + f_2(y + mx) + x f_3(y + mx)$$

$$\text{i.e., } z = f_1(y + x) + f_2(y - x) + x f_3(y - x)$$

$$P.I. = \frac{1}{D^3 + D^2 D' - DD'^2 - D'^3} e^x \cos 2y$$

$$= \text{R.P. } \frac{1}{D^3 + D^2 D' - DD'^2 - D'^3} e^x e^{i2y}$$

$$= \text{R.P. } \frac{1}{D^3 + D^2 D' - DD'^2 - D'^3} e^{x+i2y}$$

Replace D by 1 and D' by $2i$

$$= \text{R.P. } \frac{1}{(1)^3 + (1)^2(2i) - 1(2i)^2 - (2i)^3} e^{x+i2y}$$

$$= \text{R.P. } \frac{1}{1 + 2i + 4 + 8i} e^{x+i2y}$$

$$= \text{R.P. } \frac{1}{5(1+2i)} e^{x+i2y}$$

$$= \frac{1}{5} \text{R.P. } \frac{1}{(1+2i)} \frac{(1-2i)}{(1-2i)} e^{x+i2y}$$

$$= \frac{1}{5} \text{R.P. } \frac{(1-2i)}{1+4} e^x e^{i2y}$$

$$= \frac{e^x}{25} \text{ R.P. } (1 - 2i) (\cos 2y + i \sin 2y)$$

$$\therefore P.I. = \frac{e^x}{25} (\cos 2y + 2 \sin 2y).$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y+x) + f_2(y-x) + x f_3(y-x) + \frac{e^x}{25} \cos 2y + 2 \sin 2y$$

Example 1.4.24 Solve : $(D^2 - 2DD' + D'^2)z = x^2 y^2 e^{x+y}$
(Nov.'03, Apr.'11)

Solution . Given $(D^2 - 2DD' + D'^2)z = x^2 y^2 e^{x+y}$.

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 - 2m + 1 = 0$$

$$(m-1)^2 = 0 \Rightarrow m = 1, 1.$$

The complementary function (C.F.) is

$$z = f_1(y+mx) + x f_2(y+mx)$$

$$\text{i.e., } z = f_1(y+x) + x f_2(y+x)$$

$$P.I. = \frac{1}{D^2 - 2DD' + D'^2} x^2 y^2 e^{x+y} = \frac{1}{(D-D')^2} x^2 y^2 e^{x+y}$$

Replace D by $D+1$ and D' by $D'+1$ we get

$$= e^{x+y} \frac{1}{[(D+1) - (D'+1)]^2} x^2 y^2$$

$$= e^{x+y} \frac{1}{(D-D')^2} x^2 y^2$$

$$= e^{x+y} \frac{1}{D^2 \left[1 - \frac{D'}{D}\right]^2} x^2 y^2$$

$$= e^{x+y} \frac{1}{D^2} \left[1 - \frac{D'}{D}\right]^{-2} x^2 y^2$$

$$\begin{aligned}
 &= e^{x+y} \frac{1}{D^2} \left[1 + \frac{2D'}{D} + \frac{3D'^2}{D^2} + \dots \right] x^2 y^2 \\
 &= e^{x+y} \frac{1}{D^2} \left[x^2 y^2 + \frac{2D'}{D} x^2 y^2 + \frac{3D'^2}{D^2} x^2 y^2 + \dots \right] \\
 &= e^{x+y} \frac{1}{D^2} \left[x^2 y^2 + \frac{2}{D} 2y x^2 + \frac{3}{D^2} 2x^2 \right] \\
 &= e^{x+y} \left[\frac{1}{D^2} x^2 y^2 + \frac{4y}{D^3} x^2 + \frac{6}{D^4} x^2 \right] \\
 &= e^{x+y} \left[\frac{x^4 y^2}{3 \times 4} + \frac{4yx^5}{3 \times 4 \times 5} + \frac{6x^6}{3 \times 4 \times 5 \times 6} \right] \\
 \therefore P.I. &= e^{x+y} \left[\frac{x^4 y^2}{12} + \frac{yx^5}{15} + \frac{x^6}{60} \right]
 \end{aligned}$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y+x) + x f_2(y+x) + e^{x+y} \left[\frac{x^4 y^2}{12} + \frac{yx^5}{15} + \frac{x^6}{60} \right].$$

Example 1.4.25. Solve : $r + s - 6t = y \cos x$

Solution . Given equation is

$$\begin{aligned}
 r + s - 6t &= y \cos x \\
 \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} &= y \cos x \\
 (D^2 + DD' - 6D'^2)z &= y \cos x \quad \dots(1)
 \end{aligned}$$

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$\begin{aligned}
 m^2 + m - 6 &= 0 \\
 (m - 2)(m + 3) &= 0 \\
 \Rightarrow m &= 2, -3.
 \end{aligned}$$

The complementary function (C.F.) is

$$\begin{aligned}
 z &= f_1(y + m_1 x) + f_2(y + m_2 x) \\
 \text{i.e., } z &= f_1(y + 2x) + f_2(y - 3x)
 \end{aligned}$$

$$\begin{aligned}
 P.I. &= \frac{1}{D^2 + DD' - 6D'^2} y \cos x \\
 &= \text{R.P.} \frac{1}{D^2 + DD' - 6D'^2} y e^{ix} \quad \text{Replace } D \text{ by } D + i \\
 &= \text{R.P.} \frac{e^{ix}}{(D + i)^2 + (D + i)D' - 6D'^2} y \\
 &= \text{R.P.} e^{ix} \frac{1}{-1 + 2iD + D^2 + DD' + iD' - 6D'^2} y \\
 &= \text{R.P.} e^{ix} \frac{1}{-[1 - (2iD + D^2 + DD' + iD' - 6D'^2)]} y \\
 &= -\text{R.P.} e^{ix} [1 - (2iD + D^2 + DD' + iD' - 6D'^2)]^{-1} y \\
 &= -\text{R.P.} e^{ix} [1 + (2iD + D^2 + DD' + iD' - 6D'^2) + \dots] y \\
 &= -\text{R.P.} e^{ix} [y + iD'(y)] \\
 &= -\text{R.P.} (\cos x + i \sin x) [y + i]
 \end{aligned}$$

$$\therefore P.I. = -y \cos x + \sin x.$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y + 2x) + f_2(y - 3x) - y \cos x + \sin x.$$

Example 1.4.26. Solve : $(D^2 - 5DD' + 6D'^2)z = y \sin x$

Solution . Given equation is $(D^2 - 5DD' + 6D'^2)z = y \sin x \dots (1)$

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 - 5m + 6 = 0$$

$$(m - 2)(m - 3) = 0$$

$$\Rightarrow m = 2, 3.$$

The complementary function (C.F.) is

$$z = f_1(y + m_1x) + f_2(y + m_2x)$$

$$\text{i.e., } z = f_1(y + 2x) + f_2(y + 3x)$$

$$\begin{aligned}
P.I. &= \frac{1}{D^2 - 5DD' + 6D'^2} y \sin x \\
&= \text{I.P.} \frac{1}{D^2 - 5DD' + 6D'^2} y e^{ix} \\
&\quad \text{Replace } D \text{ by } D + i \\
&= \text{I.P.} \frac{e^{ix}}{(D + i)^2 - 5(D + i)D' + 6D'^2} y \\
&= \text{I.P.} e^{ix} \frac{1}{-1 + 2iD + D^2 - 5DD' - 5iD' + 6D'^2} y \\
&= \text{I.P.} e^{ix} \frac{1}{-[1 - (2iD + D^2 - 5DD' - 5iD' + 6D'^2)]} y \\
&= -\text{I.P.} e^{ix} [1 - (2iD + D^2 - 5DD' - 5iD' + 6D'^2)]^{-1} y \\
&= -\text{I.P.} e^{ix} [1 + (2iD + D^2 - 5DD' - 5iD' + 6D'^2) + \dots] y \\
&= -\text{I.P.} e^{ix} [y - 5iD'(y)] \\
&= -\text{I.P.} (\cos x + i \sin x) [y - 5i]
\end{aligned}$$

$$\therefore P.I. = 5 \cos x - y \sin x.$$

Thus the complete solution is,

$$z = \text{C.F.} + \text{P.I.}$$

$$\therefore z = f_1(y + 2x) + f_2(y + 3x) + 5 \cos x - y \sin x.$$

Example 1.4.27. Solve : $(D^2 - DD' - 2D'^2)z = (y - 1)e^x$

Solution . Given equation is $(D^2 - DD' - 2D'^2)z = (y - 1)e^x \dots (1)$

In (1), replace D by m and D' by 1 and equating to zero. Then the auxiliary equation is

$$m^2 - m - 2 = 0$$

$$(m - 2)(m + 1) = 0$$

$$\Rightarrow m = 2, -1.$$

The complementary function (C.F.) is

$$z = f_1(y + m_1x) + f_2(y + m_2x)$$

$$\text{i.e., } z = f_1(y + 2x) + f_2(y - x)$$

1.5 Part - A : Two Marks Questions

Question 1. Form the partial differential equation by eliminating the constants a and b from $z = ax + by + ab$. Nov.'13

Solution . Given equation $z = ax + by + ab$(1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = a. \Rightarrow a = p. \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = b. \Rightarrow b = q \quad \dots(3)$$

Using (2) and (3) in (1), we get $z = px + qy + pq$.

Question 2. Form the partial differential equation by eliminating the constants from $z = ax + by + a^2 + b^2$. Apr.'11

Solution . Given equation $z = ax + by + a^2 + b^2$(1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = a. \Rightarrow a = p. \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = q = b. \Rightarrow b = q \quad \dots(3)$$

Using (2) and (3) in (1), we get $z = px + qy + p^2 + q^2$.

Question 3. Form the partial differential equation by eliminating arbitrary constants from $(x - a)^2 + (y - b)^2 + z^2 = c^2$. Where c is a fixed constant.

Solution . Given equation $(x - a)^2 + (y - b)^2 + z^2 = c^2$. Nov.'11
...(1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) + 2z.p = 0. \Rightarrow (x - a) = -zp. \quad \dots(2)$$

$$2(y - b) + 2z.q = 0. \Rightarrow (y - b) = -zq. \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} (-zp)^2 + (-zq)^2 + z^2 &= c^2. \\ \Rightarrow z^2(p^2 + q^2 + 1) &= c^2. \end{aligned}$$

Question 4. Form the partial differential equation by eliminating the constants from $(x - a)^2 + (y - b)^2 + z^2 = 1$.

Nov.'12

Solution . Given equation $(x - a)^2 + (y - b)^2 + z^2 = 1$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) + 2z.p = 0. \Rightarrow (x - a) = -zp. \quad \dots(2)$$

$$2(y - b) + 2z.q = 0. \Rightarrow (y - b) = -zq. \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} (-zp)^2 + (-zq)^2 + z^2 &= 1. \\ \Rightarrow z^2(p^2 + q^2 + 1) &= 1. \end{aligned}$$

Question 5. Form the partial differential equation by eliminating the constants a and b from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$. Apr.'13

Solution . Given equation $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) = 2z \cot^2 \alpha \cdot \frac{\partial z}{\partial x}. \Rightarrow (x - a) = zp \cot^2 \alpha. \quad \dots(2)$$

$$2(y - b) = 2z \cot^2 \alpha \cdot \frac{\partial z}{\partial y}. \Rightarrow (y - b) = zq \cot^2 \alpha. \quad \dots(3)$$

Substitute (2) and (3) in (1), we get

$$\begin{aligned} z^2 p^2 \cot^4 \alpha + z^2 q^2 \cot^4 \alpha &= z^2 \cot^2 \alpha. \\ \Rightarrow p^2 + q^2 &= \tan^2 \alpha. \end{aligned}$$

Question 6. Form the partial differential equation by eliminating the arbitrary constants from $z = (x - a)^2 + (y - b)^2 + 1$. Nov.'15

Solution . Given equation $z = (x - a)^2 + (y - b)^2 + 1$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = 2(x - a). \Rightarrow (x - a) = \frac{p}{2}. \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = 2(y - b). \Rightarrow (y - b) = \frac{q}{2}. \quad \dots (3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} z &= \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 + 1 \\ \Rightarrow 4z &= p^2 + q^2 + 4. \end{aligned}$$

Question 7. Form the partial differential equation by eliminating the arbitrary constants from $z = (x - a)(y - b)$. Apr.'14

Solution . Given equation $z = (x - a)(y - b)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = 1.(y - a). \Rightarrow (y - a) = p.$$

$$\frac{\partial z}{\partial y} = q = 1.(x - b). \Rightarrow (x - b) = q.$$

Using in (1) we get $z = qp$.

Question 8. Form the partial differential equation by eliminating the arbitrary constants from $z = (x^2 + a^2)(y^2 + b^2)$. Apr.'12, Apr.'15

Solution . Given equation $z = (x^2 + a^2)(y^2 + b^2)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = 2x.(y^2 + b^2). \Rightarrow (y^2 + b^2) = \frac{p}{2x}.$$

$$\frac{\partial z}{\partial y} = q = 2y.(x^2 + a^2). \Rightarrow (x^2 + a^2) = \frac{q}{2y}.$$

Using in (1), we get $z = \left(\frac{q}{2y}\right) \left(\frac{p}{2x}\right). \Rightarrow pq = 4xyz.$

Question 9. Form the partial differential equation by eliminating the constants from $z = (x^2 + a)(y^2 + b)$.

Nov.'14

Solution . Given equation $z = (x^2 + a)(y^2 + b)$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$\begin{aligned}\frac{\partial z}{\partial x} &= p = 2x.(y^2 + b). \Rightarrow (y^2 + b) = \frac{p}{2x}. \\ \frac{\partial z}{\partial y} &= q = 2y.(x^2 + a). \Rightarrow (x^2 + a) = \frac{q}{2y}.\end{aligned}$$

$$\text{Using in (1), we get } z = \left(\frac{p}{2x}\right)\left(\frac{q}{2y}\right). \Rightarrow pq = 4xyz.$$

Question 10. Solve $\frac{\partial z}{\partial y} = \sin x$.

Apr.'11

Solution . Given PDE is $D'z = \sin x$.

AE is [Replace $D = 1$ and $D' = m$] $m = 0$. $\therefore$ CF is $z = f(x + 0y)$.

$$\text{P.I} = \frac{1}{D'} \sin x = \int \sin x \, dy = y \sin x.$$

Thus the solution is $z = f(x) + y \sin x$.

Aliter: Given equation $\frac{\partial z}{\partial y} = \sin x$. $\Rightarrow dz = \sin x \, dy$.

Integrating we get $\int dz = \sin x \int dy + f(x)$. i.e., $z = y \sin x + f(x)$.

Question 11. Form the pde by eliminating the arbitrary function from $z = f(x^2 - y^2)$. Apr.'11, Apr.'12

Solution . Given function $z = f(x^2 - y^2)$ (1)

Differentiate (1) partially w.r.to x and y , we get [where $f' = f'(x^2 - y^2)$]

$$\frac{\partial z}{\partial x} = p = f'.2x \Rightarrow f' = \frac{p}{2x}, \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = q = f'.(-2y) \Rightarrow f' = -\frac{q}{2y}. \quad \dots (3)$$

From equations (2) and (3) we have

$$\frac{p}{2x} = -\frac{q}{2y}. \Rightarrow 2yp + 2xq = 0. \quad (\text{or}) \quad yp + xq = 0.$$

Question 12. Form the partial differential equation by eliminating the arbitrary function 'f' from $f(z - xy, x^2 + y^2)$.

Nov.'10, Nov.'11, Apr.'13

Solution . Let the given function is of the form $f(u, v) = 0$ (1)

Where $u = z - xy$, $v = x^2 + y^2$.

Then $\frac{\partial u}{\partial x} = p - y$ and $\frac{\partial u}{\partial y} = q - x$ Also $\frac{\partial v}{\partial x} = 2x$ and $\frac{\partial v}{\partial y} = 2y$

Thus the elimination of f is,
$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\text{i.e., } \begin{vmatrix} p - y & 2x \\ q - x & 2y \end{vmatrix} = 0$$

$$2y(p - y) - 2x(q - x) = 0$$

$$y(p - y) - x(q - x) = 0$$

$$yp - xq = y^2 - x^2.$$

Question 13. Find the complete integral of $p + q = pq$ Nov.'11

Solution . Given equation is $p + q = pq$... (1)

It is of the form $F(p, q) = 0$.

Let the solution of (1) be $z = ax + by + c$ (2)

Partially differentiate (2) w.r.to x and y we get $p = a$ and $q = b$

Substitute $p = a$ and $q = b$ in (1) we get

$$a + b = ab. \text{ i.e., } ab - b = a. \Rightarrow b = \frac{a}{a-1} \quad \dots (3)$$

Substitute b value in (2) we get $z = ax + \frac{a}{a-1}y + c$.

Which is the complete integral.

Question 14. Solve : $xp + yq = z$.

Nov.'10, Apr.'14

Solution . Given equation is $xp + yq = z$ (1)

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} \quad \dots(2)$$

Where $P = x$, $Q = y$ and $R = z$.

Taking first and second ratio, we have

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating, $\int \frac{dx}{x} = \int \frac{dy}{y} + \log c_1$

$$\log x = \log y + \log c_1$$

$$\log \frac{x}{y} = \log c_1$$

$$\Rightarrow \frac{x}{y} = c_1. \quad \dots(3)$$

Taking second and third ratio, we have

$$\frac{dy}{y} = \frac{dz}{z}$$

Integrating, $\int \frac{dy}{y} = \int \frac{dz}{z} + \log c_2$

$$\log y = \log z + \log c_2$$

$$\log \frac{y}{z} = \log c_2$$

$$\Rightarrow \frac{y}{z} = c_2. \quad \dots(4)$$

From (3) and (4), the general solution is $\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$.

Question 15. Find complete solution of $(p + q)(z - px - qy) = 1$.

Apr.'12

Solution . Given equation $(p + q)(z - px - qy) = 1$ is written as

$$(z - px - qy) = \frac{1}{p + q}$$

$$\therefore z = px + qy + \frac{1}{p + q} \quad \dots(1)$$

This is of the Clairaut's form $z = px + qy + f(p, q)$

∴ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + \frac{1}{a+b}.$$

Question 16. Solve $(D^2 - 2DD' + D'^2)z = 0$.

Apr.'11

Solution . Given equation is, $(D^2 - 2DD' + D'^2)z = 0$.

...(1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 2m + 1 = 0$$

$$(m-1)^2 = 0. \Rightarrow m = 1, 1$$

The required solution is, $z = f_1(y+x) + xf_2(y+x)$.

Question 17. Solve $(D^2 - 4DD' + 4D'^2)z = 0$.

Nov.'11

Solution . Given equation is, $(D^2 - 4DD' + 4D'^2)z = 0$.

...(1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0. \Rightarrow m = 2, 2.$$

The required solution is, $z = f_1(y+2x) + xf_2(y+2x)$.

Question 18. Solve $(D^3 - 3DD'^2 + 2D'^3)z = 0$. Nov.'12, Apr.'15

Solution . Given equation is, $(D^3 - 3DD'^2 + 2D'^3)z = 0$.

...(1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^3 - 3m + 2 = 0$$

$$(m-1)(m^2 + m - 2) = 0$$

$$(m-1)(m+2)(m-1) = 0. \Rightarrow m = -1, -1, 2.$$

The required solution is,

$$z = f_1(y-x) + xf_2(y-x) + f_3(y+2x).$$

Question 19. Solve $(D^2 - 6DD' + 9D'^2)z = 0$.

Nov.'14

Solution . Given equation is, $(D^2 - 6DD' + 9D'^2)z = 0$.

Replace D by m and D' in (1). Then the auxiliary equation is

$$\begin{aligned} m^2 - 6m + 9 &= 0 \\ (m - 3)^2 &= 0. \Rightarrow m = 3, 3. \end{aligned}$$

The required solution is,

$$z = f_1(y + 3x) + xf_2(y + 3x).$$

Question 20. Solve : $2\frac{\partial^2 z}{\partial x^2} - 5\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 0$

Apr.'15

Solution . Given equation is written as, $(2D^2 - 5DD' + 2D'^2)z = 0$.

Auxiliary equation is [Replace D by m and D']

$$\begin{aligned} 2m^2 - 5m + 2 &= 0 \\ 2m^2 - 4m - m + 2 &= 0 \\ 2m(m - 2) - 1(m - 2) &= 0. \\ (2m - 1)(m - 2) &= 0. \Rightarrow m = \frac{1}{2}, 2 \end{aligned}$$

The required solution is,

$$z = f_1\left(y + \frac{1}{2}x\right) + f_2(y + 2x).$$

Question 21. Solve : $(D - D')^3 z = 0$.

Nov.'15

Solution . Replace $D = m$ and $D' = 1$ then the auxiliary equation is $(m - 1)^3 = 0. \Rightarrow m = 1, 1, 1$.

Thus the solution is $y = f_1(y + x) + xf_2(y + x) + x^2 f_3(y + x)$.

Question 22. Find the particular integral of

$$(D^2 + 2DD' + D'^2)z = e^{x-y}.$$

Apr.'12

Solution .

$$P.I. = \frac{1}{D^2 + 2DD' + D'^2} e^{x-y}$$

Replace D by 1 and D' by -1 we get

$$= \frac{1}{1 - 2 + 1} e^{x-y} = \frac{1}{0} e^{x-y}$$

$$= \frac{x}{2D - 2D'} e^{x-y} = \frac{x}{2 + 2} e^{x-y}$$

$$\therefore P.I. = \frac{x}{4} e^{x-y}$$

Question 23. Find the particular integral of $(D^2 + 4DD')z = e^x$
 Solution . Nov.'13

$$P.I. = \frac{1}{D^2 + 4DD'} e^{x+0y}$$

Replace D by 1 and D' by 0 we get

$$= \frac{1}{1^2 + 4 \cdot 0} e^{x+0y} = e^x$$

Question 24. Find the PI of $(D^2 + 3DD' + 2D'^2)z = \sin(x + 5y)$.
 Solution . Nov.'15

$$P.I. = \frac{1}{D^2 + 3DD' + 2D'^2} \sin(x + 5y)$$

Replace $D^2 = -1^2$, $DD' = -1 \times 5$ and $D'^2 = -5^2$ we get

$$= \frac{1}{1 + 3(-5) + 2(-25)} \sin(x + 5y)$$

$$\therefore P.I. = -\frac{1}{64} \sin(x + 5y).$$

Question 25. Find the PI of $r - 4s + 4t = e^{2x+y}$.
 Solution .

$$P.I. = \frac{1}{D^2 - 4DD' + 4D'^2} e^{2x+y} \quad \text{Replace } D = 2 \text{ and } D' = 1$$

$$= \frac{1}{4 - 8 + 4} e^{2x+y} = \frac{1}{0} e^{2x+y}$$

$$= \frac{x}{2D - 4D'} e^{2x+y} = \frac{x}{4 - 4} e^{2x+y} = \frac{x}{0} e^{2x+y}$$

$$\therefore P.I. = \frac{x^2}{2} e^{2x+y}.$$